

Table 16 Indications and contraindications for implantation of a left ventricular assist device

Indications ^a
HF with severely reduced LVEF and one of the following:
INTERMACS 1–2 with neurological and other end-organ recovery but no cardiac recovery on temporary MCS OR
INTERMACS 2–4 OR
- Severe symptoms and signs (NYHA class III/IV; INTERMACS >4) despite optimal FMT and GDIT, and at least one of the following: - Unable to exercise due to HF or, if able to perform CPET, with pVO₂ <12–14 mL/kg/min and/or <50% predicted value. - End-organ dysfunction (worsening kidney and/or hepatic function, group 2 pulmonary hypertension, cardiac cachexia) due to reduced perfusion as indicated by cardiac index ≤2 L/min/m² despite euvolaemic or therapy-resistant hypervolaemic status. - Repeated HFH within the previous 12 months without an obvious precipitating cause.
Contraindications
Absolute:
- Severe right ventricular dysfunction despite euvolaemic status - Severe irreversible end-organ (kidney and/or liver) disease - Severe irreversible neurological disease - Severe pulmonary disease - Contraindication to long-term oral anticoagulation - Medical non-adherence and/or other psychosocial limitations - Insufficient social supports to achieve compliant care in the outpatient setting - Psychological instability that jeopardizes proper follow-up and intensive therapeutic regime after implantation
Relative:
- Obesity or malnutrition - Musculoskeletal disease that impairs rehabilitation - Malignancy (depending on type and prognosis) - Severe peripheral arterial disease - Systemic infection - Alcohol or substance abuse within the past 6 months, or active smoking - Impaired cognitive function - Frailty

CPET, cardiopulmonary exercise testing; HF, heart failure; HFH, heart failure hospitalization; INTERMACS, Interagency Registry for Mechanically Assisted Circulatory Support; LVEF, left ventricular ejection fraction; MCS, mechanical circulatory support; NYHA, New York Heart Association; pVO₂, peak oxygen consumption; RHC, right heart catheterization.
^aIndication by transplantation centre includes assessment of HF severity by RHC and/or CPET.

therapy in patients with advanced HF. REMATCH showed lower all-cause death with a previous type of LVAD therapy (HeartMate XVE) compared with medical treatment. However, mortality rates at 2 years were high in both arms.⁶¹¹ Other studies have not been randomized or compared different devices or post-implant strategies.^{604,606,607,610,612,613} Prospective RCTs have been initiated

to study strategies of early LVAD implantation vs prolonged medical treatment: Early-VAD (NCT02387112),⁶¹⁴ SweVAD,⁶¹⁵ AMBU-VAD (NCT04768322), and TEAM-HF (NCT06526195).
Histological examination of heart tissue after LVAD implantation or transplantation can identify potentially treatable or genetic causes of HF and is valuable for family screening and determining potential for myocardial recovery in LVAD patients.^{616–619}

Recommendation Table 13 — Recommendations for durable mechanical circulatory support in advanced heart failure (see [Evidence Table 74](#))

Recommendations	Class ^a	Level ^b
Durable MCS (LVAD) is recommended in selected ^c patients with advanced HFrEF, despite FMT and GDIT, as BTT, BTC, BTR, or as destination therapy to improve symptoms and reduce the risk of death. ^{604,611,613,620,621}	I	C
Histopathological examination of explanted heart tissue from LVAD or heart transplantation surgery should be considered in patients with non-ischaemic HF to identify an aetiological diagnosis that may be treatable or facilitate family screening. ^{616,618}	IIa	C

BTC, bridge to candidacy; BTR, bridge to recovery; BTT, bridge to transplantation; FMT, foundational medical therapy; GDIT, guideline-directed interventional therapy; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; LVAD, left ventricular assist device; MCS, mechanical circulatory support.
^aClass of recommendation.
^bLevel of evidence.
^cEligibility criteria are reported in [Table 16](#).

8.2.3. Heart transplantation

Heart transplantation, although never evaluated in RCTs, remains the gold-standard treatment for advanced HF in the absence of contraindications. Post-transplant 1-year survival is 80%–90% and median survival is around 12.5 years.^{622,623} Transplantation significantly improves QoL and functional status. The main challenge after heart transplantation is the balance between efficacy and side effects of immunosuppression. Organ donor shortage remains the main limitation. Several countries have now started donation after circulatory death programmes, besides donation after brain death, with comparable mid-term results^{624,625} and, as a result, the numbers of heart transplantations have increased. Nevertheless, careful recipient selection remains essential, based on pre-transplant and post-transplant life expectancy (both influenced by pre-operative status and comorbidities). The main indications and contraindications for heart transplantation are listed in [Table 17](#).⁵⁸⁰
Although patients younger than 65 years of age might be more appropriate candidates due to their overall life expectancy, most programmes accept patients up to 70 years of age, and biological age as well as chronological age must be taken into account. Surgical complexity [previous sternotomies, mediastinal radiation, adult congenital heart disease (ACHD)] should also be considered. The decision pathway to transplantation or LVAD is rarely straightforward and is unique to each patient.

Table 17 Indications and contraindications for heart transplantation

Indications ^a
Advanced HF and no other therapeutic option (except for durable MCS as BTT)
Absolute contraindications
Inability to comply with the therapeutic regimen (e.g. dementia)
Serious comorbidities with poor prognosis, not reversible with heart transplantation
Other contraindications
Active infection ^b
Severe peripheral arterial or cerebrovascular disease
Elevated pulmonary vascular resistance (if pharmacologically irreversible) ^c
Cancer ^d
Irreversible liver dysfunction (cirrhosis) or irreversible kidney dysfunction ^e
Systemic disease with multiorgan involvement
BMI >35 kg/m ² or severe cachexia
Alcohol or substance abuse or active smoking within the past 6 months
Psychological instability
Insufficient social support to achieve compliant care in the outpatient setting

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BMI, body mass index; BTT, bridge to transplantation; CPET, cardiopulmonary exercise testing; HF, heart failure; LVAD, left ventricular assist device; MCS, mechanical circulatory support; RHC, right heart catheterization.

^aIndication by heart transplantation centre includes assessment of HF severity by RHC and/or CPET.

^bActive infection is a relative contraindication to transplantation, but infected LVAD may be an indication.

^cLVAD as bridge to candidacy should be considered.

^dHighly selected cases may be eligible, to be discussed with an oncologist.

^eCombined heart–liver or heart–kidney transplantation may be considered.

Eligibility for each option may change according to the specific conditions of each patient, which may also change over time. Some younger patients prefer to remain on LVAD for a prolonged period, rather than proceeding to transplantation immediately, to increase the potential number of life years gained. Other factors not related to the patient, such as average time on the heart transplantation waiting list, the centre's surgical experience, and resources, can also influence decision-making.

Recommendation Table 14 – Recommendations for heart transplantation (see [Evidence Tables 75 and 76](#))

Recommendation	Class ^a	Level ^b
Heart transplantation is recommended in selected ^c patients with advanced HF refractory to FMT and GDIT, and without contraindications, to improve QoL and survival. ^{622,626}	I	C

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FMT, foundational medical therapy; GDIT, guideline-directed interventional therapy; HF, heart failure; QoL, quality of life.

^aClass of recommendation.

^bLevel of evidence.

^cEligibility criteria are reported in [Table 17](#).

8.2.4. Symptom control and end-of-life care

Advance care planning is recommended in parallel to advanced HF treatments to determine goals of care and improve QoL along the disease trajectory.⁶²⁷ For indications and the core components of end-of-life care, see [Supplementary data online, Table S20](#).

End-of-life care in advanced HF focuses on symptom management (including breathlessness, pain, and anxiety and depression), patient comfort, and maintaining QoL ensuring personalized, compassionate care. It involves clear communication, shared decision-making, and integrating palliative or hospice care to support both the patient and their family. For further details on symptom management, see [Supplementary data online, Section S5.2.3](#).

Patients on durable MCS may request withdrawal of life-sustaining treatments because these are no longer consistent with their goals of care, because of poor QoL or newly developed serious comorbidities such as cancer. The MCS may be deactivated after careful evaluation of the patient's wish and by, or under the supervision of, a specialized team. This decision has to take national laws into account. For details regarding follow-up in patients with advanced HF see [section 11.6.2](#) and [Recommendation Table 26](#).

9. Cardiovascular comorbidities

9.1. Arrhythmias and conduction disturbances

9.1.1. Atrial fibrillation

Heart failure and AF frequently coexist, with AF occurring in more than half of patients with HF, and HF occurring in more than one-third of patients with AF. The prevalence of AF is slightly higher in patients with HFpEF compared with HFrEF.^{195,628–631}

The two conditions are deeply interrelated with shared risk factors and pathophysiological mechanisms. The coexistence of HF and AF is associated with a high risk of death, HFH, and stroke regardless of LVEF.^{195,628–631} HF induced by AF appears to have a more favourable outcome than HF causing AF.^{13,632}

The management of patients with HF and AF include identification of triggers, prevention of embolic events, management of HF, and rate or rhythm control of AF.

9.1.1.1. Identification of triggers and risk factor management

Precipitating factors for AF – e.g. hyperthyroidism, toxic agents (e.g. including alcohol), valvular heart diseases, infection, diabetes, uncontrolled hypertension, obesity, sleep disorders, and volume overload – should be routinely identified and appropriately managed. Diuretic therapy is indicated to relieve congestion and may increase the chances of restoring sinus rhythm by reducing filling pressures and ventricular rate.

9.1.1.2. Prevention of embolic events

Oral anticoagulation is recommended in patients with HF and AF to reduce the risk of thromboembolism, according to the CHA₂DS₂-VA score ([Table 18](#)).^{118,633–635} However, patients with HCM and CA are at higher risk of thromboembolism irrespective of CHA₂DS₂-VA score, and anticoagulation is recommended for all these patients with AF.^{636–642} In the absence of mechanical prosthetic heart valves and/or moderate

Table 18 Updated definitions for the CHA₂DS₂-VA score

	Description	Point
C	Chronic HF stage B, C, D	1
H	Hypertension	1
A	Age ≥75 years	2
D	Diabetes mellitus	1
S	Stroke/transient ischaemic attack/arterial thromboembolism	2
V	Vascular disease (CAD and/or PAD)	1
A	Age 65–74 years	1

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CAD, coronary artery disease; CHA₂DS₂-VA, congestive heart failure, hypertension, age ≥75 years (2 points), diabetes mellitus, prior stroke/transient ischaemic attack/arterial thromboembolism (2 points), vascular disease, age 65–74 years; PAD, peripheral artery disease.
A CHA₂DS₂-VA score of 2 or more is recommended as an indicator of elevated thromboembolic risk for decisions on initiating oral anticoagulation.
A CHA₂DS₂-VA score of 1 should be considered an indicator of elevated thromboembolic risk for decisions on initiating oral anticoagulation.

or severe mitral stenosis, direct-acting oral anticoagulants are recommended as the anticoagulant of choice.^{643–646} Left atrial appendage closure can be an option for selected patients as described in the *2024 ESC Guidelines for the management of atrial fibrillation*.^{118,647–651}

9.1.1.3. Management of heart failure and atrial fibrillation

All patients with coexisting HF and AF should be optimally treated with FMT, AMT, and GDIT regardless of AF (Section 6.2.4).⁴³⁵
In patients with AF and HFrEF, where there may be a high likelihood of LVEF recovery following rhythm control, evidence remains limited. However, given the well-established prognostic benefit of FMT on long-term prognosis in the general HFrEF population, initiation of FMT upfront seems reasonable.
There is ongoing debate regarding the optimal management of AF in patients with HF. Currently, there is insufficient evidence to support a rhythm-control strategy over a rate-control strategy.^{337,652–658} Decisions with regard to choice of strategy must be made taking into account the cause–effect relationship between AF and HF, along with clinical presentation, symptom burden, stage of disease, comorbidities, and patient preference.

9.1.1.4. Rate control

Data to support strict vs lenient rate control in patients with HF and AF are inconclusive. Analyses from a large observational registry showed a significant association between heart rate in AF and long-term death only in patients with HFrEF and heart rate >100 b.p.m.^{659,660} The RACE II trial demonstrated that lenient rate control (target heart rate <110 b.p.m.) was non-inferior to strict rate control (<80 b.p.m. at rest; <110 b.p.m. during exercise) in patients with permanent AF with a primary endpoint of death from CV causes, HFH, stroke, systemic embolism, bleeding, and life-threatening arrhythmic events.³⁰⁷ A post-hoc combined analysis from the AFFIRM and the RACE trials showed similar results.⁶⁶¹ However, these trials were not focused on HF and only a minority of patients included had HF.
A large observational study showed that a lenient rate-control strategy and poor rate control are associated with a higher risk

of death as compared with strict rate control in patients with AF and HF regardless of LVEF.⁶⁶² Lenient rate control, with resting heart rate <110 b.p.m., should be considered as initial target in both patients with and without HF, with re-evaluation based on symptoms.¹¹⁸
Beta-blockers have proven safety and efficacy in patients with AF and HFrEF and are recommended as the first-line approach.^{663,664} If the ventricular rate control is suboptimal with beta-blockers, or if beta-blockers are not tolerated, digoxin or diltiazem should be considered.^{665–667} In patients with HFrEF who are haemodynamically unstable, i.v. amiodarone or digoxin may be used to achieve acute control of heart rate and improve haemodynamics, although acute electrical cardioversion to restore sinus rhythm should be first choice.^{668–671} There is no evidence on pharmacological strategies for rate control in patients with AF and HFpEF.
Atrioventricular node ablation combined with CRT was found to be superior to pharmacological therapy in reducing death and HFH and improving QoL in patients with severely symptomatic permanent AF and HF, regardless of LVEF.^{672,673}

9.1.1.5. Rhythm control

The results of RCTs comparing catheter ablation vs pharmacological rhythm control in patients with HFrEF cannot be considered conclusive for hard outcomes on HFH and death, although catheter ablation has proven superior to antiarrhythmic treatment for AF recurrence.^{674–682} Data on the role of catheter ablation in HFpEF are limited.^{683,684} For a detailed description, see [Supplementary data online, Section S6.1.1](#).^{674–681,683–686} Based on current evidence, catheter ablation should be considered only in carefully selected patients with HFrEF and symptomatic AF. This Task Force suggests that these criteria should be used to guide the selection of patients for catheter ablation: high burden AF,⁶⁸⁷ less than 1-year duration of continuous persistent AF, and a clear cause–effect relationship between AF and HF.
In patients presenting with DHF, AF with rapid ventricular rates, and haemodynamic instability, urgent cardioversion is recommended. Pharmacological and electrical cardioversion seem to have similar outcomes, although specific studies in HF populations are lacking.^{688,689} Amiodarone is recommended in patients with HFrEF requiring pharmacological cardioversion. Other antiarrhythmic drugs, i.e. propafenone, flecainide, and dronedarone, are associated with poorer outcomes in HFrEF and should be avoided.^{690–694} There are no data on pharmacological cardioversion in HFpEF.

Recommendation Table 15 – Recommendations for management of atrial fibrillation in patients with heart failure (see *Evidence Tables 77–81*)

Recommendations	Class ^a	Level ^b
Recommendations for anticoagulation		
Oral anticoagulation is recommended in patients with clinical AF at elevated thromboembolic risk, as determined by CHA ₂ DS ₂ -VA score ^c , to prevent ischaemic stroke and thromboembolism. ⁶³⁵	I	A

Continued

DOACs are recommended in preference to VKAs in patients with HF ^a to prevent stroke and thromboembolism. ^{643–646}	I	B1
Rate control		
Beta-blockers are recommended in stable patients with HFrEF and AF as first-line therapy for short- and long-term rate control. ^{663,665}	I	C
Digoxin should be considered in stable patients with HFrEF and AF when the ventricular rate remains high despite beta-blockers, or when beta-blockers are contraindicated/not tolerated, in order to obtain short- and long-term rate control. ^{665–667}	IIa	C
Atrioventricular node ablation combined with CRT should be considered in patients with severely symptomatic permanent AF and poor rate control despite medical therapy and at least one HFH to reduce symptoms, physical limitations, recurrent HFH, and death. ^{672,673}	IIa	B1
Intravenous amiodarone or digoxin may be considered in haemodynamically unstable patients with HFrEF and AF to stabilize the patient and achieve acute control of heart rate. ^{668–671}	IIb	C
Cardioversion		
Urgent cardioversion is recommended in the setting of DHF in patients presenting with rapid ventricular rates and haemodynamic instability to restore sinus rhythm. ^{669,671,692,695}	I	C
Atrial fibrillation catheter ablation		
Catheter ablation for AF should be considered in selected ^e patients with symptomatic AF and HFrEF to improve QoL and reduce the risk of HFH or death. ^{674–676,678,681,682,684,696}	IIa	C

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AF, atrial fibrillation; CHA₂DS₂-VA, congestive heart failure, hypertension, age ≥75 years (2 points), diabetes mellitus, prior stroke/transient ischaemic attack/arterial thromboembolism (2 points), vascular disease, age 65–74 years; CRT, cardiac resynchronization therapy; DHF, decompensated heart failure; DOAC, direct-acting oral anticoagulant; HF, heart failure; HFH, heart failure hospitalization; HFrEF, heart failure with reduced ejection fraction; QoL, quality of life; VKA, vitamin K antagonist.

^aClass of recommendation.

^bLevel of evidence.

^cSee Table 18.

^dExcept for patients with moderate or severe mitral stenosis and in patients with mechanical prosthetic heart valves where VKAs are recommended.

^eAll the following criteria are required: high burden AF, less than 1-year duration for continuous persistent AF, clear cause–effect relationship between AF and HF.

9.1.2. Ventricular arrhythmias

Ventricular arrhythmias may cause HF or be a consequence thereof. In sustained ventricular tachycardia, initial management should include stabilization (electrical cardioversion), followed by identification and correction of precipitating factors, and optimization of FMT. Although amiodarone reduces ventricular arrhythmias, it does not reduce SCD or death.⁶⁹⁷ Patients with HF and resuscitated cardiac arrest or life-threatening ventricular arrhythmias have an indication for a secondary prevention ICD.¹¹⁹

Heart failure can occur because of frequent PVCs.⁶⁹⁸ A burden of PVCs of about 10% precipitates an increased risk of HF and the risk increases with higher burden.^{119,699–701} Thus, regular assessment of LVEF in this setting is indicated. Amiodarone may be considered to reduce recurrent arrhythmias and improve LV function,⁶⁹⁷ with regular follow-up to assess possible side effects. Catheter ablation of PVCs may improve LV function and, possibly, outcomes in patients with PVCs.^{702–705} For further details, please refer to the 2022 ESC Guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death.¹¹⁹

9.1.3. Bradyarrhythmias

Pacing indications in patients with HF do not differ from those without HF. However, long-term RV pacing can adversely impact LV function.⁷⁰⁶ In patients with HFrEF, where there is an expectation of high pacing requirements, i.e. complete heart block or AF with slow ventricular response, CRT implantation should be considered to improve outcomes (see Recommendation Table 7).⁴⁰⁹

9.2. Chronic coronary syndromes

Obstructive CAD is a frequent cause of HF.³³ Coronary microvascular dysfunction affects up to 80% of patients with HFpEF and seems to be associated with poor prognosis.^{707–709} Diagnostic work-up of chronic coronary syndromes in patients with HF is described in Section 5. Treatment of chronic coronary syndrom in patients with established HF is described below.

9.2.1. Medical therapy

Beta-blockers are recommended as first-line therapy in patients with HFrEF and CAD due to their prognostic effects, closely followed by other FMT drugs.^{288,290,291,710–712} Ivabradine should be considered in patients with HFrEF in sinus rhythm with heart rate >70 b.p.m. despite beta-blockers at the highest tolerated dose or if beta-blockers are contraindicated.^{294,713} Other medications such as amlodipine,⁷¹⁴ felodipine,⁷¹⁵ nicorandil,⁷¹⁶ and ranolazine⁷¹⁷ do not have effects on death but may improve symptoms. Trimetazidine may improve LV function and exercise capacity in patients with HFrEF and chronic coronary syndrome already on beta-blockers.^{718–720} Importantly, initiation and/or uptitration of antianginal therapy should not lead to reduction in FMT. Short-acting nitrates can cause hypotension and should be used with caution. Non-dihydropyridine calcium channel blockers (verapamil and diltiazem) are not recommended in patients with HFrEF due to their negative inotropic effects which may precipitate the risk of HF events.⁷²¹ The use of beta-blockers or calcium channel blockers do not appear to be associated with worse outcomes in HFpEF.^{722,723} Treatment of patients with HFpEF should be similar to that of patients without HF (see the 2024 ESC Guidelines for the management of chronic coronary syndromes).¹⁰

9.2.2. Myocardial revascularization

Data to support the prognostic benefit of myocardial revascularization in patients with stable obstructive CAD and HF are limited. Two RCTs compared revascularization vs medical therapy in patients with LVEF ≤35% and multivessel coronary disease.^{178–180} The STICH trial compared coronary artery bypass grafting (CABG) to medical therapy and showed survival benefit

for CABG at long-term follow-up.^{179,180} The REVIVED-BCIS2 study compared percutaneous coronary intervention (PCI) and optimal medical therapy to optimal medical therapy alone and was neutral.¹⁷⁸ The patient populations differed between the two trials and patients in REVIVED-BCIS2 were 10 years older, making comparison between them difficult. For details of these results and other studies, see [Supplementary data online, Section S6.2.1](#).^{178,180,724–735}

Patients with HF and chronic coronary syndrome due to obstructive CAD must receive FMT before revascularization. A multidisciplinary Heart Team discussion including an HF specialist is recommended to decide whether conservative treatment or an invasive treatment strategy should be adopted. If

revascularization is planned, the multidisciplinary Heart Team should provide an individualized recommendation on the revascularization strategy—CABG or PCI—depending on patient characteristics. In case of persistent angina despite FMT for HF and antianginal drugs, revascularization is recommended regardless of LVEF. If the Heart Team agrees that revascularization is the best treatment option, in patients with HFrEF, LVEF $\leq 35\%$ and multivessel disease, CABG should be the first choice ([Figure 19](#)).

The Task Force recognizes an apparent discrepancy between the latter recommendation (class of recommendation IIa, level of evidence B1) and other recommendations in this context across other guidelines (e.g. *2024 ESC Guidelines for the management of chronic coronary syndromes*, class of recommendation I, level

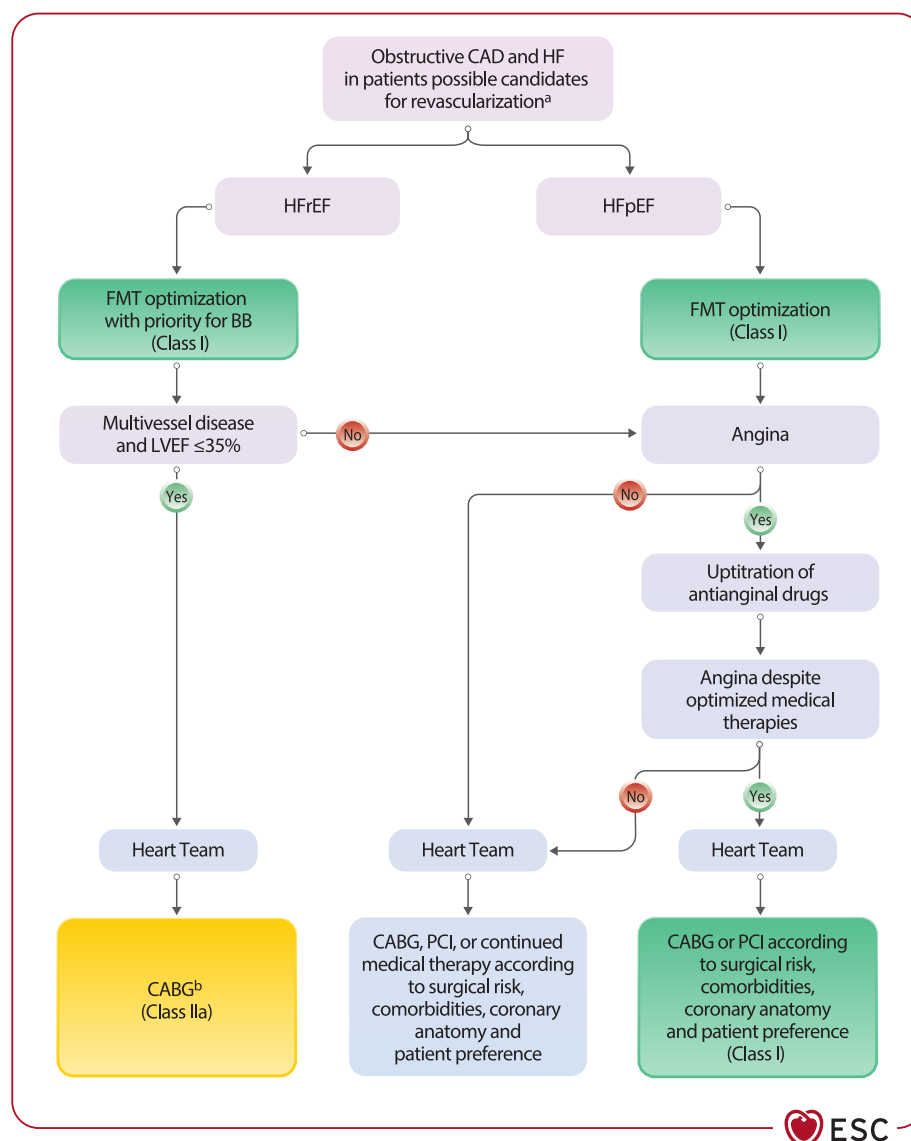


Figure 19 Management of coronary artery disease in patients with heart failure. BB, beta-blocker; CABG, coronary artery bypass grafting; CAD, coronary artery disease; FMT, foundational medical therapy; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LVEF, left ventricular ejection fraction; PCI, percutaneous coronary intervention. Colours refer to class of recommendation. ^aExclusion of frailty, severe and multiple comorbidities and limited life expectancy making the risk of revascularization higher than the expected benefit. ^bPCI or continued medical therapy are alternative treatments if the patient is not eligible for CABG.

of evidence B). However, the patient populations that were considered are slightly different in the chronic coronary syndrome guidelines (often patients with chronic coronary syndrome presenting with reduced LVEF) as compared to these guidelines (often patients with HFrEF presenting with CAD). In addition, according to the most recent system for level of evidence grading, proposed by the ESC, the Task Force considered STICHES, with all its limitations (mean age 60 years, limited number of events at 10 years, limited use of modern FMT), as the only possible study available to support the recommendation.¹⁸⁰

Recommendation Table 16 – Recommendations for revascularization in chronic coronary syndrome in patients with heart failure (see [Evidence Table 82](#))

Recommendations	Class ^a	Level ^b
Revascularization for symptom improvement (HFpEF and HFrEF)		
Coronary revascularization is recommended in patients with HF, obstructive CAD, and persistent angina, despite FMT for HF and antianginal drugs, to relieve symptoms. ^{734,735}	I	C
Revascularization for outcome improvement (HFrEF)		
If revascularization is planned after assessment by a Heart Team, CABG should be considered as the preferred revascularization strategy, in patients with LVEF ≤35% and multivessel CAD suitable for surgery, to improve long-term survival. ¹⁸⁰	IIa	B1

CABG, coronary artery bypass grafting; CAD, coronary artery disease; FMT, foundational medical therapy; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LVEF, left ventricular ejection fraction.

^aClass of recommendation.

^bLevel of evidence.

9.3. Valvular heart disease

9.3.1. Aortic stenosis

Aortic stenosis can contribute to or cause HF by increasing LV afterload.⁷³⁶ Prevalence of aortic stenosis ranges between 3% and 7% in patients with chronic HF and is higher in patients with HFpEF as compared with those with HFrEF.⁷³⁷ About 50% of patients with severe symptomatic aortic stenosis undergoing transcatheter aortic valve implantation (TAVI) have HF.⁷³⁸ Once HF symptoms develop, prognosis is poor and timely valve intervention is mandatory. HF therapies may be used cautiously and should not delay valve intervention. After valve intervention, a clinical and echocardiographic reassessment is important to detect residual HF or risk of HF readmission. In this regard, further optimization of FMT as well as treatment of comorbidities and management of procedure-related factors is appropriate.⁷³⁹ In the recent open-label DapaTAVI trial, 1222 patients with history of HF and at least one risk factor (diabetes, CKD, LV dysfunction) were randomly assigned to receive either dapa-gliflozin or standard of care. At 1-year follow-up, the risk of the primary endpoint (death from any cause or WHF events) was reduced by 28% in the treatment arm as compared with the

control arm, mainly driven by WHF events.⁷⁴⁰ Therefore, treatment with a SGLT2-I is reasonable in these patients.

Aortic valve intervention is recommended in all patients with HF and a life expectancy >1 year with high-gradient severe aortic stenosis, irrespective of LVEF, as well as in patients with severe low-flow low-gradient aortic stenosis and HFpEF. In patients with severe low-flow low-gradient aortic stenosis and HFpEF, aortic valve intervention should be considered. Details on diagnostic work-up, specific indications and the modality of intervention (TAVI vs surgery) are reported in the *2025 ESC/EACTS Guidelines for the management of valvular heart disease*.⁹

9.3.2. Aortic regurgitation

Severe aortic regurgitation can lead to LV dilatation and dysfunction which may progress to HF. Treatment with ACE-Is/ARNIs/ARBs/MRAs while awaiting aortic valve intervention can be beneficial.⁷⁴¹ Beta-blockers can prolong diastole and should be used only after careful consideration of risks and benefits.

Specific recommendations for intervention are reported in the *2025 ESC/EACTS Guidelines for the management of valvular heart disease*.⁹

9.3.3. Mitral regurgitation

9.3.3.1. Primary mitral regurgitation

Primary MR is caused by abnormalities of the valve apparatus and can lead to HF. In severe primary MR with HF symptoms, surgical repair is preferred over replacement whenever feasible.^{739,742,743} Mitral transcatheter edge-to-edge repair (TEER) should be considered in patients at high surgical risk.^{9,744,745}

9.3.3.2. Secondary mitral regurgitation

Secondary or functional MR is primarily a disease of the left cardiac chambers rather than the valvular apparatus. It is common in patients with HF, especially those with HFrEF in whom the disease has a prevalence of 25%.^{746,747} In order to adequately assess the severity of MR, the patient should be euvolaemic or as close to euvolaemia as possible. The underlying aetiology and severity of MR needs to be assessed by cardiologists specialized in echocardiography and structural heart diseases.⁷⁴⁸

A new classification, differentiating between atrial secondary MR and ventricular secondary MR has been recently proposed.⁹ Atrial secondary MR can be associated with HFpEF.⁷⁴⁹ Indications for mitral valve treatment in atrial secondary MR are reported in the *2025 ESC/EACTS Guidelines for the management of valvular heart disease*.⁹

In patients with severe ventricular secondary MR and HFrEF, optimization of FMT and GDIT with CRT implantation (if indicated) is mandatory before considering any mitral valve intervention ([Figure 20](#)). In case of concomitant CAD requiring revascularization, specific recommendations on management are reported in the *2025 ESC/EACTS Guidelines for the management of valvular heart disease*.⁹ In selected patients without need for coronary revascularization, TEER is recommended. [Table 19](#) provides the clinical and echocardiographic criteria for patient selection, which are based on inclusion criteria from RCTs. For the results of these studies, see [Supplementary data online, Section S6.3.1](#).^{750–754} If the criteria reported in [Table 19](#) are not fulfilled, other options can be considered. Specifically, in

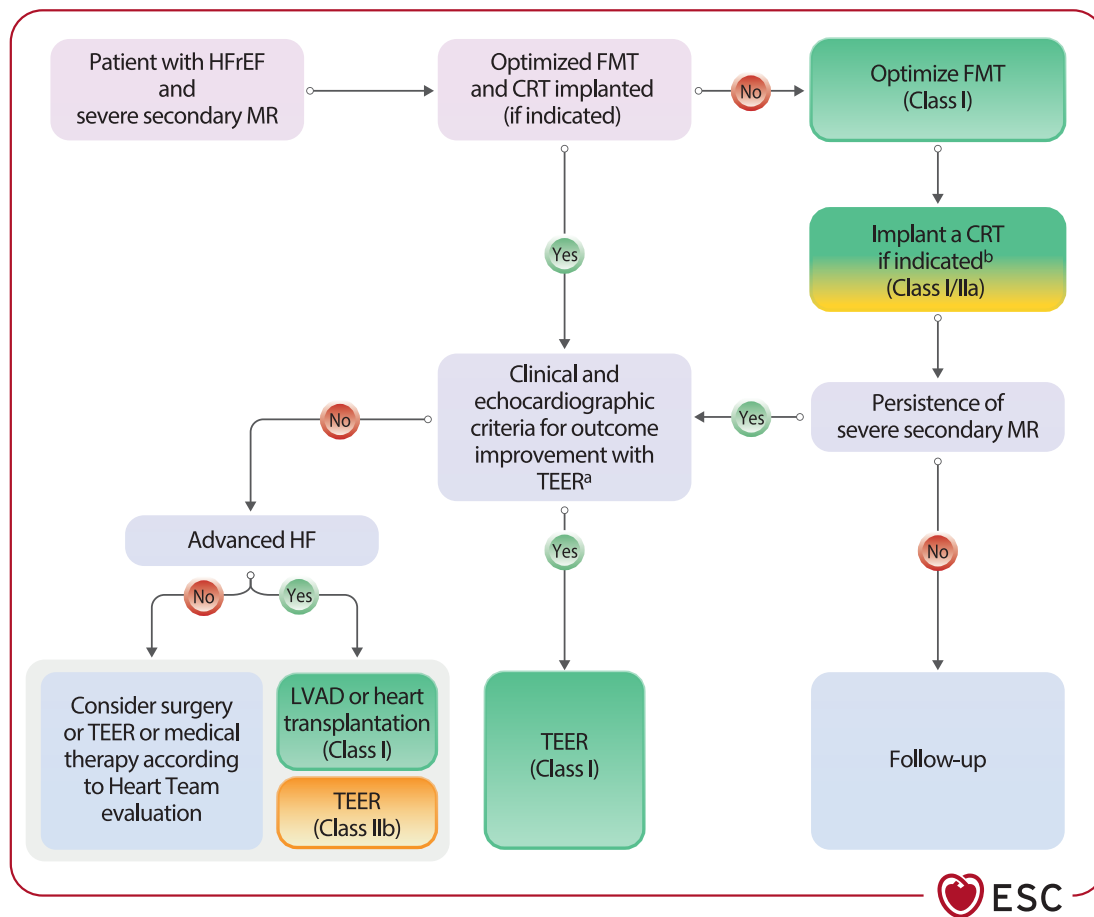


Figure 20 Management of secondary mitral regurgitation in patients with heart failure with reduced ejection fraction. CRT, cardiac resynchronization therapy; FMT, foundational medical therapy; HF, heart failure; LVAD, left ventricular assist device; MR, mitral regurgitation; TEER, transcatheter edge-to-edge repair. Colours refer to class of recommendation. ^aSee Table 19. ^bSee Figure 10.

Table 19 Selection criteria for mitral transcatheter edge-to-edge repair in heart failure with reduced ejection fraction

Clinical and echocardiographic criteria
Optimized FMT and other GDIT (i.e. CRT) if needed
NYHA class $\geq$ II
At least one HFH within the last year
OR
Increased natriuretic peptide levels (BNP $\geq$ 300 pg/mL or NT-proBNP $\geq$ 1000 pg/mL)
No stage D (advanced) HF
No CAD requiring revascularization
No hypertrophic, restrictive, or infiltrative cardiomyopathies
LVEF 20%–50%
LVESD $\leq$ 70 mm
Anatomy deemed suitable for mitral TEER
sPAP $\leq$ 70 mmHg
No severe RV dysfunction
No severe aortic and/or tricuspid valve disease

BNP, B-type natriuretic peptide; CAD, coronary artery disease; CRT, cardiac resynchronization therapy; FMT, foundational medical therapy; GDIT, guideline-directed interventional therapy; HF, heart failure; HFH, heart failure hospitalization; LVEF, left ventricular ejection fraction; LVESD, left ventricular end-systolic diameter; TEER, transcatheter edge-to-edge repair; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; RV, right ventricular; sPAP, systolic pulmonary artery pressure.

patients with advanced HF, LVAD or heart transplantation is recommended as first choice and TEER may be considered for symptom improvement in patients not suitable or waiting for these therapies (see Section 8) (Figure 20). Notably, patients with ventricular secondary MR experience a higher residual risk of death or HFH despite TEER.^{755,756} Thus, close follow-up in HF clinics is warranted.⁷⁵⁷

Recommendation Table 17 – Recommendations for valvular heart disease in patients with heart failure (see Evidence Tables 83 and 84)

Recommendations	Class ^a	Level ^b
Aortic stenosis		
Aortic valve intervention, SAVR or TAVI, is recommended in patients with severe aortic stenosis and HF ^c to improve symptoms and reduce the risk of death. ^{758–765}	I	B1
Secondary mitral regurgitation		
Mitral transcatheter edge-to-edge repair is recommended in haemodynamically stable, symptomatic patients with HFrEF and persistent severe secondary mitral regurgitation despite optimized FMT and CRT if indicated, who fulfil specific clinical and echocardiographic criteria, ^d in order to reduce the risk of HFH and improve QoL. ^{750–753}	I	B1
Mitral transcatheter edge-to-edge repair may be considered in selected symptomatic patients with HFrEF and persistent severe secondary mitral regurgitation despite optimized FMT and CRT if indicated (after excluding the option of LVAD or heart transplantation) who do not fulfil the clinical and echocardiographic criteria for outcome improvement, ^d in order to improve symptoms. ^{766–769}	IIb	C

CRT, cardiac resynchronization therapy; FMT, foundational medical therapy; HF, heart failure; HFH, heart failure hospitalization; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LVAD, left ventricular assist device; LVEF, left ventricular ejection fraction; QoL, quality of life; SAVR, surgical aortic valve replacement; TAVI, transcatheter aortic valve implantation.

^aClass of recommendation.

^bLevel of evidence.

^cThis recommendation refers to patients with severe high gradient aortic stenosis regardless of LVEF and to patients with low-flow low-gradient aortic stenosis and HFrEF. For details regarding low-flow low-gradient aortic stenosis in HFpEF, we refer to the 2025 ESC/EACTS Guidelines for the management of valvular heart disease.⁹

^dFor clinical and echocardiographic criteria, see Table 19.

9.3.4. Tricuspid regurgitation

Severe TR has a high prevalence (9%–20%) in both patients with HFpEF and HFrEF.^{747,770}

Tricuspid regurgitation is currently classified as primary TR, CIED-related TR, and secondary TR. Secondary TR is most common.^{747,770} Medical therapy to target TR is limited.^{747,771} The development of transcatheter devices to treat TR provides an alternative to surgery in selected patients. Three RCTs are

currently available that investigated the role of transcatheter treatments in patients with severe symptomatic TR. They showed a significant benefit in terms of QoL in patients receiving device therapy as compared with those treated conservatively, but without reduction of death or HFH at 1-year follow-up.^{772–774} Recently, a possible benefit in terms of HFH was reported in patients treated with tricuspid TEER, but with a high rate of crossover after 1-year follow-up.^{775,776} It should be noted that all these studies were unblinded and did not focus on HFpEF or HFrEF populations.

Indications for TR treatment are reported in the 2025 ESC/EACTS Guidelines for the management of valvular heart disease.⁹

9.4. Hypertension

Hypertension is common in patients with HF, especially those with HFpEF.^{454,777,778} In hypertensive heart disease, LV remodelling and LV hypertrophy lead to both diastolic and systolic impairment. Specific blood pressure targets for patients with hypertension and HF have not been evaluated in RCTs.

In patients presenting with HF due to refractory hypertension, secondary causes [i.e. primary aldosteronism, renovascular hypertension, and obstructive sleep apnoea (OSA)] should be ruled out and treated.⁶²

Indications for the treatment of hypertension in HF are reported in the 2024 ESC Guidelines for the management of elevated blood pressure and hypertension.⁶²

Lifestyle interventions, such as increasing physical activity, optimizing weight, reducing salt and alcohol intake, and smoking cessation, are beneficial in all cases.⁶²

The optimal treatment strategy for hypertension in patients with HFpEF is uncertain. FMT with effect on blood pressure (i.e. MRA) should be started, continued, or uptitrated. Loop diuretics reduce congestion and decrease blood pressure. In addition, ACE-Is, ARBs, and calcium channel blockers cause LV regression and therefore may be beneficial in both controlling blood pressure and improving HFpEF symptoms.⁶² Patients with HFpEF have limited preload reserve; therefore, normotension should be maintained and hypotension avoided.

In patients with HFrEF, treatment strategies are similar to those for normotensive patients, as FMT also lowers blood pressure. Once FMT has been initiated and uptitrated, suboptimal blood pressure control is rare in HFrEF. If additional agents are required, amlodipine or felodipine may be considered provided the patient is euvolaemic.^{714,715} Alpha-blockers may be continued for the treatment of prostatic hyperplasia in the absence of hypotension but are not recommended for the treatment of hypertension in HFrEF. Non-dihydropyridine calcium channel blockers, e.g. verapamil and diltiazem, and central sympathetic agonists, e.g. moxonidine, are contraindicated due to poor outcomes in patients with HFrEF.⁷⁷⁹ In patients with HFrEF, asymptomatic hypotension should be accepted in order to maintain target doses of FMT.⁷⁸⁰

9.5. Stroke

Heart failure and stroke share common risk factors. Approximately 10%–24% of patients with ischaemic stroke have concomitant HF either *de novo* or pre-existing.^{781–784} Conversely, patients with HF have an increased risk of stroke compared with those without.⁷⁸⁵ A history of stroke is reported

in 8%–11% patients with HF^{785–789} with similar^{790–794} or increased^{795,796} prevalence in patients with HFpEF compared with HFrEF.

Poorer outcomes after stroke are observed in patients with HF with a 2.2–4.5 times higher rate of death compared with people without HF,^{782,783,797} as well as an increased rate of recurrent stroke.⁷⁹⁸

The risk of stroke in patients with HF seems independent of rhythm, with an annualized stroke risk of 2.0 per 100 patient-years in patients in sinus rhythm and 2.87 per 100 patient-years in patients with AF.^{799–802} However, RCTs failed to demonstrate a possible benefit of oral anticoagulation in patients with HFrEF in sinus rhythm.^{803–806} Thus, in the absence of AF, there is no evidence to recommend routine anticoagulation for stroke prevention in patients with HFrEF.

Optimization of blood pressure, heart rate, and cardiac output immediately after a stroke is key to reducing the risk of DHF and improving outcomes. In patients already established on FMT, treatment with FMT should be maintained if possible and attention should be paid to dose modification in order to avoid both hypo- and hypertension as well as rebound tachycardia in patients in whom beta-blockers have been withdrawn.⁸⁰⁷ Following the acute phase, treatment of HF in patients with stroke does not differ from treatment in patients without stroke.

10. Non-cardiovascular comorbidities

10.1. Obesity

Obesity contributes to the development of well-known CV and HF risk factors like hypertension, diabetes, and dyslipidaemia. Obesity has direct adverse effects on cardiac structure and function, and is associated with the development of CVD, including HF.⁸⁰⁸ Patients with HF and obesity may have lower natriuretic peptide concentrations due to increased expression of clearance receptors, augmented peptide degradation by the adipose tissue,¹⁵⁵ and underestimation of circulatory congestion.⁸⁰⁹ Peak oxygen consumption adjusted for body weight underestimates exercise capacity in obese patients, and an adjustment for lean body mass should be used for risk stratification pro-inflammatory.⁸¹⁰

A stronger association between obesity and HFpEF rather than HFrEF has been described.^{778,811} Among patients with HFpEF, the obesity-related phenotype has a high prevalence (30%–40% in Europe and 60%–70% in North America).^{314,812} These patients display several pathophysiological mechanisms that differ from non-obese patients with HFpEF.⁸¹³ An increase in adipocyte mass induces a state of systemic inflammation and a proinflammatory transformation of epicardial adipose tissue that may lead to fibrosis. The capacity of the left ventricle to dilate in response to an increase in blood volume is impaired, and thus, even modest degrees of volume overload may lead to cardiac overfilling and disproportionate increase in cardiac filling pressures.⁸¹⁴ On average, patients with HFpEF and obesity are younger, experience worse HF signs and symptoms, poorer QoL, and worse functional capacity, than those without obesity.^{815,816}

Weight-loss interventions in patients with HFpEF and obesity alleviate symptoms and may improve outcomes. Lifestyle interventions, such as caloric restriction, exercise, and patient education, have a significant impact on body weight and are recommended in all patients with HF.^{817,818} Recent RCTs

demonstrated favourable effects of the GLP-1 RA semaglutide and the dual GIP/GLP-1 RA tirzepatide in patients with HFpEF and obesity regardless of diabetes status, namely reduction of body weight, improvement of QoL, symptoms, and functional capacity (see [Supplementary data online, Section S7.1](#)).^{315–317} Pooled and secondary analyses of RCTs, and the recent SUMMIT trial, have also suggested a potential benefit of these drugs on HF events.^{317,819,820}

Dedicated RCTs are needed to evaluate safety and efficacy of these drugs in patients with HFrEF and to assess long-term follow-up.^{821,822}

In propensity score-matched cohort studies, bariatric surgery was associated with weight loss and a lower risk of all-cause and CV death, HFH, and AF as compared to medical therapy in patients with HF and a BMI ≥35 kg/m², with consistent benefits observed among patients with HFrEF and HFpEF.^{823,824}

Studies comparing bariatric surgery vs new anti-obesity medications are lacking, and the indications for bariatric surgery may change in the future.

Recommendation Table 18 – Recommendations for obesity in patients with heart failure (see Evidence Tables 85 and 86)

Recommendations	Class ^a	Level ^b
Semaglutide or tirzepatide should be considered for patients with symptomatic HF, LVEF ≥45%, and BMI ≥30 kg/m ² , regardless of diabetes status, to reduce body weight, and improve exercise capacity and QoL. ^{315–317,819,820,825}	IIa	B1
Bariatric surgery may be considered in obese patients with HF and a BMI ≥35 kg/m ² to reduce body weight when repetitive and structured lifestyle changes combined with weight-reducing medications do not result in maintained weight loss. ^{823,824}	IIb	C

BMI, body mass index; HF, heart failure; LVEF, left ventricular ejection fraction; QoL, quality of life.
^aClass of recommendation.
^bLevel of evidence.

10.2. Diabetes

Patients with HF who have T2DM have a poorer prognosis than patients who do not have T2DM.^{826–828}

The diagnostic process and therapeutic approach to HF in patients with T2DM is the same as the population without T2DM. There is no evidence that the efficacy of FMT is altered by the presence of T2DM.^{241,829–833 60} The 2023 ESC Guidelines for the management of cardiovascular disease in patients with diabetes dedicated a specific section to the treatment of diabetes in patients with HF.⁶⁰

Sodium–glucose co-transporter 2 inhibitors have been investigated in different populations with diabetes, ranging from patients with ASCVD or multiple ASCVD risk factors to patients with established HF across the entire range of LVEF or recently hospitalized for WHF, with greater absolute risk reduction of clinical events in patients at higher risk.⁸³⁴ Thus, the use of

SGLT2-Is is recommended in all patients with HF and T2DM, independently of glycated haemoglobin levels or other concomitant glucose-lowering agents. SGLT2-Is are generally well tolerated, although they may cause urinary infections and genital fungal infections as vulvovaginitis and balanitis with a prevalence of up to 6.9% in women and 4.8% in men. They are not indicated in patients with type 1 diabetes mellitus due to the risk of diabetic ketoacidosis (reported incidence of 2%–3%).^{835,836} Semaglutide or tirzepatide should be added if not contraindicated in patients with HFpEF and T2DM, especially if they are also obese.^{316,317}

Metformin, despite its long history as first-line treatment of T2DM, has no dedicated CV outcome trials to assess its safety or efficacy; observational studies reported a lower risk of death or HFH in patients with HF when compared with insulin and sulfonylureas.^{837–839} Metformin is not recommended in patients with an eGFR <30 mL/min/1.73 m² or hepatic impairment because of the risk of lactic acidosis.

In the context of T2DM, insulin glargine and insulin degludec have been evaluated in specific CV outcome trials, demonstrating a neutral effect on incident HF in patients with diabetes.^{840–842} In post-hoc analyses of RCTs and retrospective registries, use of insulin was associated with poorer outcomes among patients with HF. Thus, if insulin is needed in a patient with HF, patients should be monitored for WHF after initiation of treatment.^{843,844}

The dipeptidyl peptidase-4 inhibitor saxagliptin and thiazolidinediones (glitazones) have been associated with an approximately 30% increased risk of HFH and, thus, are contraindicated in patients with HF.^{845,846} No consistent data are available for the use of sulphonylureas in patients with HF.⁶⁰

10.3. Chronic kidney disease

Chronic kidney disease and HF frequently coexist.⁸⁴⁷ They share common risk factors and may develop simultaneously. CKD may lead to worsening of cardiac function and vice versa. The incidence of CKD is five-fold higher in patients with HF as compared with those without.⁸⁴⁸ About 30%–40% of individuals with chronic HF have CKD, with the highest proportions, up to 50%, in HFpEF.^{778,849,850} CKD is a major independent determinant of increased mortality and morbidity in HF.⁸⁵⁰ Albuminuria and low GFR are independently associated with an increased risk of major ASCVD, HF events, and CV death.^{61,851–854} The presence of CKD might have an impact on FMT optimization, and it is often cited as a reason for FMT under-utilization in clinical practice.⁸⁵⁵ Even in most patients with CKD the initiation of beta-blockers, ACE-Is/ARNIs/ARBs, SGLT2-Is, and MRAs is recommended.¹⁶⁴

Several RCTs have shown that patients with HF and concomitant CKD are at higher risk of events as compared to those without CKD. However, no interaction between drug effects and kidney function was noted in subgroup analyses of these trials.^{242–245,309} Furthermore, the beneficial effects of FMT are similar, or even higher, irrespective of CKD.^{251,252,847,856–858}

An evidence gap remains for the treatment of HF in patients with severe CKD as these patients were generally excluded from RCTs.

Worsening kidney function in patients with chronic HF is not necessarily associated with worse outcomes. For example, after initiation of ACE-Is/ARNIs/ARBs, MRAs or SGLT2-Is, an initial decrease in the glomerular filtration pressure may decrease GFR and increase serum creatinine. These changes are generally transient and occur despite improvement in patient outcomes and may delay long term deterioration of kidney function.^{243,250–253} Sacubitril-valsartan, compared with enalapril, led to a slower decline in kidney function, despite a slight increase in the UACR, and improved CV outcomes to a similar extent in patients with CKD vs other patients in the PARADIGM-HF trial.²⁷⁹ In a participant-level pooled analysis of the DAPA-HF and DELIVER trials, patients experiencing worsening of eGFR to <25 mL/min/1.73 m² were at heightened risk for development of subsequent CV events. Continuation of dapagliflozin was associated with lower rates of the primary composite outcome.⁸⁵⁹

A prespecified secondary analysis of the FINEARTS-HF trial, in patients with HF and LVEF ≥40%, reported an acute decline in eGFR of −2.9 mL/min/1.73 m² (95% CI −3.4 to −2.4) during the first 3 months on finerenone vs placebo.⁸⁶⁰ However, finerenone still reduced the risk of new-onset micro- and macroalbuminuria by 24% and 38%, respectively.⁸⁶⁰

Thus, a transient decrease in kidney function after the initiation of ACE-Is/ARNIs/ARBs and MRAs or SGLT2-Is should not prompt their interruption. An increase in serum creatinine of <50% above baseline, as long as eGFR remains >15 mL/min/1.73 m², is considered acceptable.

Notably, in patients with HFpEF, semaglutide and tirzepatide may acutely reduce eGFR; however, they exhibit a long-term protective effect on kidney function.^{861,862}

Chronic kidney disease is highly prevalent in patients with DHF. Worsening or improvement of kidney function occurs commonly, affecting about 30%–50% of patients admitted with HF.²⁷³ Adverse CV outcomes are observed in patients with worsening kidney function and poor diuretic response, but not in patients with good diuretic response.^{506,863}

Data from the STRONG-HF trial showed that a decrease in eGFR within 1 week after discharge was associated with more evidence of congestion and with an increased incidence of HF readmissions.⁸⁶⁴ Conversely, worsening kidney function is not associated with poor outcomes in the context of successful decongestion.^{506,863} For further details refer to the 2026 ESC Guidelines on cardiovascular disease and chronic kidney disease.¹²²

10.4. Iron deficiency and anaemia

Iron deficiency and anaemia are common in patients with HF and are associated with reduced functional capacity, and worse QoL and prognosis independently of each other.^{778,865–868} Iron deficiency is slightly more prevalent in patients with HFpEF than in patients with HFrEF.^{865,869}

In patients with HF, ID has been defined as either ferritin <100 ng/mL, or ferritin 100–299 ng/mL with TSAT <20%. This definition has been adopted by the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure³ and the 2023 Focused Update of the 2021 ESC Guidelines for the

diagnosis and treatment of acute and chronic heart failure,⁵ and has been used in almost all recent trials assessing the effects of i.v. iron supplementation in patients with HF and ID. However, it has recently been recognized that in patients with HF this definition does not reliably identify patients with absolute or functional ID. The current definition of ID may include patients with TSATs ($\geq 20\%$) and serum ferritin levels (20–100 mg/L) in the normal range, who are probably not iron deficient, exhibit a better prognosis, and might not respond to i.v. iron therapy.⁸⁷⁰ Moreover, serum ferritin levels are highly sensitive to inflammation and oxidative stress. Thus, an alternative definition of ID—TSAT $< 20\%$ —has been proposed, mainly based on evidence from explorative analyses and meta-analyses of RCTs. A TSAT $< 20\%$ has higher sensitivity and specificity compared with the previous definition of ID and is associated with higher mortality rates, independently of HF phenotype.^{869–871} Moreover, some meta-analyses of large RCTs reported that lower TSAT values better identify patients that will respond favourably to i.v. iron supplementation.^{872–874} The Task Force supports the use of this most recent definition of ID in patients with HF and recognizes the need for new trials investigating the efficacy of i.v. iron supplementation in patients with HF and ID as defined by TSAT $< 20\%$.

Several trials, including AFFIRM-AHF,⁴⁹⁶ CONFIRM-HF,⁸⁷⁵ FAIR-HF,^{876,877} EFFECT-HF,⁸⁷⁸ and IRONMAN⁸⁷⁹ (where ID was defined as either ferritin < 100 ng/mL, or ferritin 100–299 ng/mL with TSAT $< 20\%$), have demonstrated that iron supplementation with i.v. ferric carboxymaltose is safe and effective in improving symptoms and QoL in patients with HF and LVEF $< 40\%$ – 45% .

Thus, i.v. iron supplementation is recommended in patients with symptomatic HFrEF and ID to alleviate HF symptoms and improve QoL. Data in patients with HFpEF are scarce.⁸⁸⁰

The prognostic benefit of i.v. iron supplementation in patients with HF has been investigated in several RCTs and meta-analyses (see [Supplementary data online, Section S7.2](#)).^{496,872,873,879,881–886} Given the conflicting results of major RCTs, the position of the Task Force is that evidence regarding the benefits of i.v. iron supplementation on strong endpoints is not conclusive. This is primarily due to the use of the old definition of ID as an enrolment criterion in almost all trials, the heterogeneity of RCTs in terms of enrolled populations as well as of i.v. iron doses and timing of administration and that some major RCTs were affected by the coronavirus disease 2019 pandemic. Thus, i.v. iron supplementation with ferric carboxymaltose or ferric derisomaltose should be considered in patients with symptomatic HFrEF and ID to reduce the risk of HFH, taking into account that patients with TSAT $< 20\%$ might benefit most. Further trials using TSAT $< 20\%$, or other criteria that do not rely on ferritin levels as inclusion criteria, are warranted.

In patients with HFrEF and mild to moderate anaemia enrolled in the RED-HF trial,⁸⁸⁷ darbepoetin-alpha failed to reduce all-cause death or HFH and showed an increased risk of thromboembolic events. Thus, erythropoietin-stimulating agents should be avoided for the treatment of anaemia related to HF.

Recommendation Table 19 — Recommendations for iron deficiency in patients with heart failure (see [Evidence Tables 87 and 88](#))

Recommendations	Class ^a	Level ^b
Intravenous iron supplementation is recommended in patients with symptomatic HFrEF and iron deficiency to alleviate HF symptoms and improve QoL. ^{875–879}	I	B1
Intravenous iron supplementation should be considered in patients with symptomatic HFrEF and iron deficiency to reduce the risk of HFH. ^{872–874,879,881,885,886,888}	IIa	B1

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HF, heart failure; HFH, heart failure hospitalization; HFrEF, heart failure with reduced ejection fraction; QoL, quality of life.

^aClass of recommendation.

^bLevel of evidence.

10.5. Cancer

In patients with cancer, HF may occur as a result of the interactions between cancer therapy, cancer itself, and/or patients' CV backgrounds (risk factors and coexisting CVD). Current cancer treatments may cause HF through direct cardiotoxic effects, or indirectly, favouring the occurrence of pathological conditions predisposing to HF, such as volume overload from i.v. infusions and corticosteroids, myocarditis, myocardial ischaemia, systemic or pulmonary hypertension, thromboembolism, arrhythmias, or valve disease.^{6,889,890}

The definition of symptomatic and asymptomatic cancer therapy-related cardiac dysfunction is reported in [Supplementary data online, Table S21](#). Cancer drugs causing HF are reported in [Supplementary data online, Table S22](#).

In patients who develop symptomatic HF during cancer therapy, FMT is recommended, regardless of LVEF.⁶ Discontinuation vs continuation of cardiotoxic drugs should be discussed by a multi-disciplinary team. Severity of cancer therapy-related cardiac dysfunction guides the decision.^{6,891} Initiation of FMT for HFrEF is recommended in patients with asymptomatic moderate-to-severe cancer therapy-related cardiac dysfunction and should be considered in asymptomatic patients with mild presentations.

For details regarding CV toxicity risk assessment process, diagnostic work-up, follow-up procedures, and therapeutic management of cardiotoxicity, we refer to the 2022 ESC Guidelines on cardio-oncology.⁶

10.6. Lung disease and sleep-disordered breathing

Chronic obstructive pulmonary disease is common among patients with HF.^{892–894} Due to the overlap in symptoms and signs, the differentiation between HF and COPD may be difficult.⁸⁹⁵

Chronic obstructive pulmonary disease significantly impacts on symptoms and QoL, has been associated with HF undertreatment, and negatively influences prognosis of HF.^{893,896}

Treatment of HF is generally well tolerated in COPD. The use of cardioselective beta-blockers (bisoprolol, metoprolol succinate, or nebivolol) should be preferred. In patients with asthma,

these medications require more caution and should be started at low doses after consideration of the risks and benefits, with close monitoring for signs of airway obstruction, and appropriate supervision.⁸⁹⁷ Chronic obstructive pulmonary disease exacerbations should not be a cause of beta-blocker discontinuation.⁸⁹⁸ Inhaled corticosteroids and beta-adrenergic agonists have not been specifically studied in patients with HF; however, they do not seem to increase CV risk.^{899,900} In addition, appropriate management of COPD can improve cardiac function.⁹⁰¹

Sleep-disordered breathing is found in over one-third of patients with HF and is even more prevalent in DHF. Sleep-disordered breathing includes OSA, central sleep apnoea (CSA), and mixed disease. SDB negatively impacts disease progression and prognosis.^{902,903} Patients with a suspicion of SDB should be referred for overnight polysomnography to definitively diagnose SDB, and to document the predominant type of sleep apnoea (central vs obstructive).⁹⁰² The therapeutic approach to SDB includes the correction of underlying causes, when identified.

The use of adaptive servo-ventilation (ASV) in patients with HFrEF and predominantly CSA is not recommended, based on the results of the SERVE-HF trial,⁹⁰⁴ which showed an increase in both all-cause and CV death in patients with chronic HF, LVEF $\leq 45\%$, and SDB with predominant CSA.⁹⁰⁴

The ADVENT-HF trial randomized 731 patients with HF, LVEF $\leq 45\%$, and coexisting SDB, mostly OSA (73%) to standard optimal treatment alone or standard optimal treatment and ASV (1:1). Over a mean follow-up period of 3.6 years, ASV had a neutral effect on the primary composite outcome (the cumulative incidence of all-cause death, first CV hospitalization, new-onset AF/flutter, and delivery of an appropriate ICD shock).⁹⁰⁵ However, ASV improved sleep quality and structure, health-related QoL, and symptoms of patients with SDB.⁹⁰⁵

Thus, in patients with SDB and predominant OSA, nocturnal hypoxaemia can be treated with nocturnal oxygen supplementation, continuous positive airway pressure (CPAP), bilevel positive airway pressure, and ASV.

When SDB is caused by CSA, implantable phrenic nerve stimulation was found to be safe and effective in reducing the apnoea-hypopnoea index in a small RCT.^{906–908}

Recommendation Table 20 – Recommendations for lung disease and sleep-disordered breathing (see [Evidence Table 89](#))

Recommendations	Class ^a	Level ^b
Adaptive servo-ventilation may be considered in patients with HFrEF and sleep-disordered breathing with predominant obstructive sleep apnoea to improve sleep quality, health-related QoL and symptoms. ⁹⁰⁵	IIb	C
Adaptive servo-ventilation is not recommended in patients with HFrEF and sleep-disordered breathing with predominant central sleep apnoea because of an increased risk of CV and all-cause death. ⁹⁰⁴	III	A

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CV, cardiovascular; HFrEF, heart failure with reduced ejection fraction; QoL, quality of life.

^aClass of recommendation.

^bLevel of evidence.

10.7. Anxiety, depression, and cognitive impairment

Anxiety and depression are common findings in patients affected by chronic and disabling conditions, such as HF, and need to be promptly identified to minimize psychological distress and its negative consequences.⁹⁰⁹ Depression affects 20% of patients with HF, and it may affect adherence to FMT. Its prevalence is higher in women, and it is associated with worse clinical status and poorer prognosis.^{910–912}

The diagnosis of depression typically relies on standardized questionnaires and clinical interviews. Symptom overlap between HF and depression can complicate diagnosis, as both conditions may present with cognitive issues (e.g. difficulty concentrating) and physical symptoms (e.g. fatigue). Several validated questionnaires are available for screening clinically significant depressive symptoms⁹¹¹ (see [Supplementary data online, Section S7.4, Table S23](#)).

Cognitive impairment in these patients has been reported.^{911,913} However, there is still no consensus on the best therapy for patients with HF and depression.

Cognitive behavioural therapy has demonstrated significant favourable long-term (4–9 months) reduction of anxiety and depression and improvement in QoL compared with non-treated patients.^{914,915} The safety of sertraline and escitalopram has been demonstrated, although without a significant reduction in depression compared with placebo.^{916,917} Tricyclic antidepressants should be avoided for the treatment of depression in HF as they may cause hypotension, WHF, and arrhythmias.^{911,912}

10.8. Frailty

Frailty is defined as a multidimensional, dynamic, and potentially reversible state, related to, but independent of, age, that makes individuals more vulnerable to the effects of stressors and at increased risk of poor outcomes. A score assessing frailty including four domains (clinical, functional, psychocognitive, and social) has recently been proposed.⁹¹⁸

The presence of frailty might have an impact on disease management and is associated with higher rates of death and hospitalizations, and reduced health-related QoL, regardless of which tool was used to assess frailty.^{919–923} In patients with HF, frailty is particularly prevalent in older adults. When frailty is suspected or identified, a multidisciplinary approach to facilitate an individualized management is warranted.^{918,922,924}

Recommendation Table 21 – Recommendations for anxiety, depression, and frailty (see [Evidence Tables 90 and 91](#))

Recommendation	Class ^a	Level ^b
Assessment of anxiety, depression and frailty should be considered in patients with HF to support the development of personalized care plans and to identify factors that may contribute to adverse outcomes. ^{909,914,915,920,923}	IIa	C

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HF, heart failure.

^aClass of recommendation.

^bLevel of evidence.

10.9. Other non-cardiovascular comorbidities

Infections may cause WHF and can precipitate HFH.^{925,926} Infections should ideally be prevented or, if present, promptly recognized and treated to avoid disease progression.^{927–930}

Influenza vaccination is associated with a reduced risk of pneumonia and all-cause death in patients with HF.^{931–936} Thus, influenza and pneumococcal vaccination should be considered in all patients with HF in order to reduce the risk of infection-related complications.⁹³⁵

Thyroid dysfunction can cause changes in cardiac function and structure and may precipitate DHF.^{937,938} Assessment of thyroid function is recommended in patients with *de novo* HF or DHF and should be regularly assessed during follow-up of patients with HF. Treatment of thyroid dysfunction should be guided by endocrine guidelines.^{939–942}

Hyperuricaemia and consequent gout attacks can be frequent in patients with HF, mainly caused or precipitated by diuretic treatment.^{943,944} Allopurinol is recommended as the first choice to chronically reduce uric acid levels in patients with HF,⁹⁴⁵ whereas colchicine can be used for acute gout attacks. Non-steroidal anti-inflammatory drugs and selective cyclooxygenase-2 inhibitors are not recommended in patients with HF because they may precipitate decompensation and worsen kidney function.^{942,946–948}

Heart failure and erectile dysfunction frequently coexist.^{949,950} Cardioselective beta-blockers should be preferred in patients with HF and erectile dysfunction, mainly bisoprolol and nebivolol, started at the lowest doses.⁸⁹⁷ In the presence of erectile dysfunction symptoms, phosphodiesterase-V inhibitors are generally safe and can be considered in patients who are compensated,⁹⁵¹ avoiding their concomitant use with nitrates.⁹⁵²

Recommendation Table 22 – Recommendations for other non-cardiovascular comorbidities (see Evidence Tables 92 and 93)

Recommendations	Class ^a	Level ^b
Influenza and pneumococcal vaccination should be considered in patients with HF to reduce the risk of pneumonia complications, hospitalization, and death. ^{931–934,936,953,954}	IIa	C
NSAIDs and COX-2 inhibitors are not recommended in patients with HF as they increase the risk of HF worsening and HFH. ^{942,946–948}	III	B2

COX-2, cyclooxygenase-2; HF, heart failure; HFH, heart failure hospitalization; NSAID, non-steroidal anti-inflammatory drug.
^aClass of recommendation.
^bLevel of evidence.

11. Multidisciplinary team management, long-term follow-up, and patient care

11.1. Multidisciplinary management of heart failure

Multidisciplinary care for HF combines the expertise of various professionals to provide integrated, patient-centred care that

improves both clinical outcomes and QoL for patients (Figure 21).

Access to healthcare professionals with awareness of HF and expertise in its diagnosis and management is essential for early detection and better outcomes. Multidisciplinary HF care involves generalist and specialist healthcare professionals,⁹⁵⁵ including cardiologists, cardiothoracic surgeons, specialist nurses, pharmacists, and physiologists (Figure 21) (see Supplementary data online, Section S8.1).^{17,956–963}

Management by an HF multidisciplinary team (HF-MDT) across outpatient and inpatient settings improves outcomes for people with HF.^{962,964–967}

The literature evaluates various methods of delivering HF management programmes enabling local HF services to design services tailored to their HF-MDT and the structure of their healthcare system. A meta-analysis of 16 RCTs with 3999 eligible patients evaluating the benefits of multidisciplinary HF clinics compared with usual care demonstrated a reduction in HFH.⁹⁶⁷ A meta-analysis of eight RCTs with 1764 patients indicated that multidisciplinary teams may have the potential to reduce death, although a definitive RCT is needed.⁹⁶⁴ The benefits were more evident in patients with a recent emergency department presentation or HFH.⁹⁶⁶ Other methods of delivery, including home visits and transitional care models, are given in Supplementary data online, Section S8.1.^{145,962,965,968–972}

Healthcare professionals involved in the management of HF, whether specialist or generalist, should be equipped with the knowledge required to effectively assess and manage HF through appropriate education and training.^{958,973,974} Malnutrition has been associated with worse prognosis in patients with HF.⁷³⁸ Thus, nutritional assessment and dietary interventions can be useful.^{588,589} No evidence supports fluid restriction in chronic HF.^{975,976}

Recommendation Table 23 – Recommendations for the multidisciplinary management of heart failure (see Evidence Tables 94–96)

Recommendations	Class ^a	Level ^b
A multidisciplinary HF management programme is recommended in patients with HF to reduce the risk of HFH and death. ^{964–967}	I	B1
HF education and self-management strategies in patients with HF are recommended to reduce the risk of HFH or death. ⁹⁷⁷	I	A
Multidisciplinary interventions to improve adherence, including assessment of health literacy and polypharmacy review, are recommended in patients with HF to reduce the risk of hospitalization and death. ^{978–981}	I	B2

HF, heart failure; HFH, heart failure hospitalization.
^aClass of recommendation.
^bLevel of evidence.

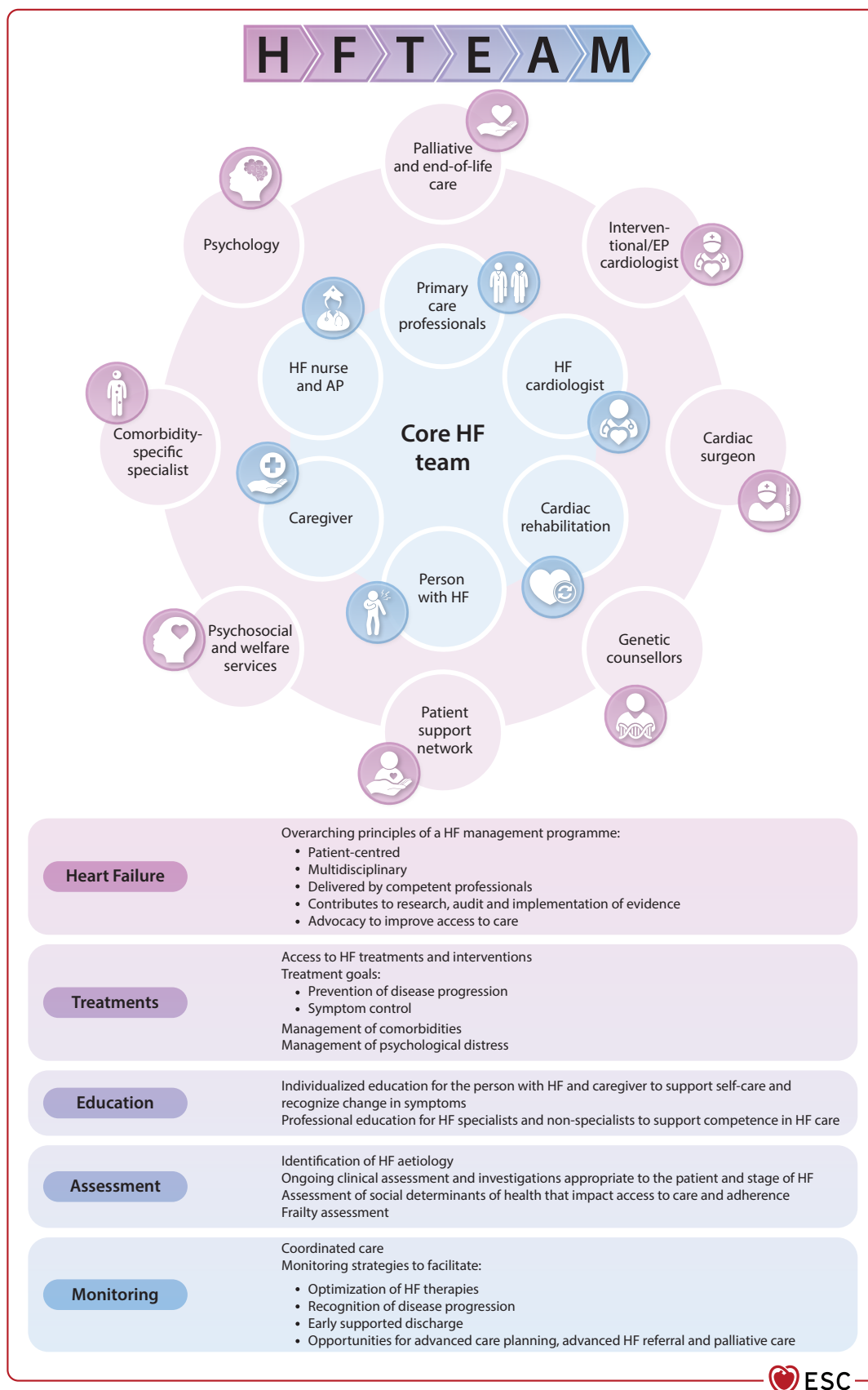


Figure 21 Core members of the heart failure team with expanded team members depending on patient needs and local availability. AP, allied professional; EP, electrophysiology; HF, heart failure.

11.2. Patient education, self-care, and lifestyle advice

Education, self-care, and lifestyle advice empower patients and their caregivers in managing HF and engage in shared decision-making. Self-care incorporates education and strategies to support maintenance, monitoring, and management of HF.⁹⁷³ The aim is to ensure people with HF and their caregivers are equipped with knowledge and skills and be partners in their care and not simply recipients of information and treatments. The benefits of self-care are described in a meta-analysis of 20 RCTs including 5624 patients.⁹⁷⁷ Self-management interventions on the background of usual HF care improved QoL and reduced time to HFH or all-cause death. There may be some barriers to self-care, but these can be overcome by careful design of programmes^{982,983} (see [Supplementary data online, Section S8.2](#)).

Effective self-care enhances QoL, reduces symptom severity, and lowers the risk of hospitalization. It is essential that HF professionals collaborate with patients and caregivers, using shared decision-making to develop personalized self-care plans ([Supplementary data online, Table S24](#)).

11.3. Implementation and adherence to therapy

11.3.1. Implementation and optimization

Implementation of HF therapies should be individualized, based on available evidence, and involve the HF-MDT and patients and caregivers. Yet, there remain delays to implementation and optimization of treatments.^{984–986} Multivariable adjusted large observational studies suggest that among the strongest factors associated with improved FMT, AMT, and GDIT use is access to cardiologist specialists.^{987–990} For barriers and facilitators to improved implementation of HF therapies, including clinical inertia, impact of race, ethnicity, gender, and health literacy, see [Supplementary data online, Table S25](#).

Evidence supports the role of HF nurses and pharmacists in the implementation and optimization of HF therapies and avoidance of unnecessary side effects of polypharmacy.^{991–994}

Pharmacists play a key role in ensuring safe and effective use of medicines, including medication reconciliation and the optimization of therapies for comorbidities. Development of the HF-MDT to support the implementation and optimization of HF and to address the barriers that exist is key to improving outcomes for patients.

11.3.2. Long-term adherence

Adherence to self-care strategies and therapies is essential to control symptoms, limit disease progression, and reduce hospitalization and death.⁹⁸¹ Factors that impact adherence and affect clinical outcomes include health literacy, polypharmacy, and managing multiple health conditions.^{978,979,995} Strategies to improve adherence in the context of HF have been described elsewhere and provide guidance on the use of education tailored to the individual, communication skills, polypharmacy review, and co-ordinated care.^{980,996} Central to these is the importance of the HF-MDT recognizing the barriers to adherence, assessing those barriers, and utilizing other members of the health and social care team to address them.

11.4. Exercise training and cardiac rehabilitation

11.4.1. Benefits of exercise training according to heart failure stage and phenotype

There is consistent evidence that exercise training improves exercise tolerance and health-related QoL in patients with HF.⁹⁴ HF-ACTION was the largest RCT on exercise training in HF including 2331 patients with LVEF $\leq 35\%$ (NYHA classes II–IV). Compared with usual care, supervised exercise training plus usual care showed modest, non-significant reductions in all-cause death and hospitalizations. However, prespecified adjustments revealed significant reductions in CV death and HFH.⁹⁹⁷ A systematic review of 60 trials with 8728 patients (mean age 63 years, mean LVEF 32%) confirmed improvements in functional capacity, QoL, and reductions in HFH with exercise training but found no reduction in death.⁹⁹⁸ Similar findings were reported by another systematic review focusing on HFrEF (LVEF $< 40\%$).^{94,999}

In HFpEF, exercise training is safe and significantly enhances exercise capacity and QoL, with no significant impact on diastolic function, arterial stiffness, or vascular function, and insufficient data exist to assess its impact on clinical events, such as death and hospital admissions, due to small sample sizes and short follow-up times.^{1000,1001}

Most trials in HF, except for HF-ACTION, are too small to reliably evaluate the effects of exercise rehabilitation across subgroups (see [Supplementary data online, Section S8.4](#)).^{1002,1003}

In patients with long-term mechanical support, the benefit of exercise training on functional capacity has been demonstrated.^{1004,1005} The Rehab-VAD trial showed that moderate intensity aerobic training was safe in patients with continuous-flow LVADs and yielded significantly improved health status, treadmill time, and leg strength, without significant improvements in peak oxygen consumption.¹⁰⁰⁶ If a patient with advanced HF can tolerate physical activity, exercise training can improve QoL.¹⁰⁰⁷ Exercise training before and after heart transplantation should follow recommendations according to the consensus document.¹⁰⁰⁸

11.4.2. Safety and contraindications

No major adverse effects of exercise training were reported in the included studies of above systematic reviews and meta-analyses.^{1001,1009} However, exercise interventions should only be initiated in stable individuals after initiation of FMT.

Exercise training is contraindicated in patients with refractory/unstable angina and other high-risk CV conditions (e.g. high-grade arrhythmias, DHF, active thromboembolic disease) and in any other unstable non-cardiac condition (e.g. active infection, uncontrolled diabetes, end-stage cancer, COPD exacerbation).⁹⁴

11.4.3. Exercise prescription

A personalized training programme, with the mode of delivery, type, and intensity tailored to the patient's preferences, risk level, and comorbidities, is recommended to improve uptake of exercise-based cardiac rehabilitation. Exercise training should be started as part of cardiac rehabilitation and be individually prescribed according to the FITT model (frequency, intensity,

time, type) for aerobic training, resistance training, and inspiratory muscle training^{1010,1011} (see [Supplementary data online, Section S8.4](#)).^{998,1012–1016}

Adherence to the prescribed exercise regimen is crucial (see [Supplementary data online, Section S8.4](#)).^{1017–1020}

Recommendation Table 24 – Recommendations for exercise training and cardiac rehabilitation (see [Evidence Tables 97–99](#))

Recommendations	Class ^a	Level ^b
Personalized exercise training, in the context of multidisciplinary exercise-based cardiac rehabilitation, is recommended for all stable patients, unless there are specific contraindications, in order to improve exercise capacity and QoL, and reduce the risk of all-cause hospitalization. ¹⁰²¹	I	B1
Personalized exercise training outside exercise-based cardiac rehabilitation is recommended for all stable patients on a long-term basis, unless there are specific contraindications, in order to improve exercise capacity and QoL, and reduce the risk of all-cause hospitalization. ¹⁰²²	I	B2
Cardiac rehabilitation ^c is recommended in patients with an LVAD to improve functional capacity and QoL. ^{1005,1023}	I	B1
Cardiac rehabilitation ^c should be considered in patients with DHF, as soon as safely possible, to improve functional capacity and QoL. ^{1002,1024–1026}	IIa	B1

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DHF, decompensated heart failure; LVAD, left ventricular assist device; QoL, quality of life.

^aClass of recommendation.

^bLevel of evidence.

^cEither comprehensive or exercise-based.

11.5. Telemonitoring

Advances in digital technology and telemonitoring have, in recent years, contributed substantially to the transformation of (remote) care delivery, from a hospital-centred approach to a patient-oriented approach. Telemonitoring can involve non-invasive and invasive monitoring modalities. Non-invasive telemonitoring relies on traditional vital parameters measured remotely (e.g. weight, heart rate, and blood pressure), which can be used to guide patients remotely to adjust therapy or seek advice either through telephone consultations or eHealth-based environments. Recent systematic reviews and meta-analyses report significant reductions in all-cause death and HFH with non-invasive telemonitoring.^{1027–1029} However, despite a growing body of evidence, results from individual trials are inconsistent and limited mainly by considerable heterogeneity in terms of patient population, modality, and intervention (see [Supplementary data online, Section S8.5](#)).^{1027–1031} At present, non-invasive telemonitoring may be considered for patients with HF in order to reduce the risk of HFH.

Device-based monitoring modalities (pacemaker/ICD/CRT) based upon thoracic impedance, heart sounds, and breathing patterns, aim to assist in the selection between stable patients and those that need rapid access to care.^{1032,1033} Many implanted devices can provide remote information either on the device itself (generator and lead function), arrhythmias, or on patient physiology (heart rate, heart sounds, bio-impedance). There is evidence that monitoring can detect device malfunction/integrity and that it may be useful for detecting arrhythmias. However, there is currently little evidence that device monitoring improves clinical outcomes, although ongoing RCTs are awaited.^{1029,1033–1039}

Invasive monitoring modalities consist of several specifically designed telemonitoring sensor technologies measuring remote haemodynamic parameters. Currently, only pulmonary artery sensor-guided therapy has been studied in multiple RCTs and has shown a benefit in reducing HFH in patients in NYHA class III and a HFH within the past 12 months.^{1029,1040–1047} Additionally, several prospective studies have confirmed the safety and efficacy of pulmonary artery-guided therapy in reducing HFH in multiple settings.^{1037,1048–1052} Remote haemodynamic monitoring by pulmonary artery-guided therapy should be considered in selected patients with moderate-to-severe HF with NYHA class III and a HFH within the past 12 months in order to reduce the risk of HFH.

Recommendation Table 25 – Recommendations for invasive and non-invasive telemonitoring (see [Evidence Table 100](#))

Recommendations	Class ^a	Level ^b
Remote haemodynamic monitoring of PA pressure should be considered in symptomatic patients with HF, NYHA class III, and an HFH in the past 12 months in order to reduce the risk of HFH. ^{1029,1043,1045}	IIa	B1
Non-invasive telemonitoring modalities may be considered for patients with HF in order to reduce the risk of HFH. ^{1027,1029,1053}	IIb	B1

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HF, heart failure; HFH, heart failure hospitalization; NYHA, New York Heart Association; PA, pulmonary artery.

^aClass of recommendation.

^bLevel of evidence.

11.6. Long-term follow-up

11.6.1. Follow-up

Patients with HF, even when symptoms are stable and well controlled, require regular in-person follow-up to ensure ongoing optimization of therapy, detect asymptomatic disease progression or comorbidities, and discuss new advances in care. For follow-up intervals of patients with a recent episode of DHF undergoing optimization or uptitration of FMT, see [Section 7](#).

For symptomatic patients with HF with no recent DHF, these guidelines recommend assessment of symptom severity, pharmacological adherence, arrhythmias, changes in the ECG and, if deemed relevant, echocardiography and laboratory parameters (including full blood count, electrolytes, iron

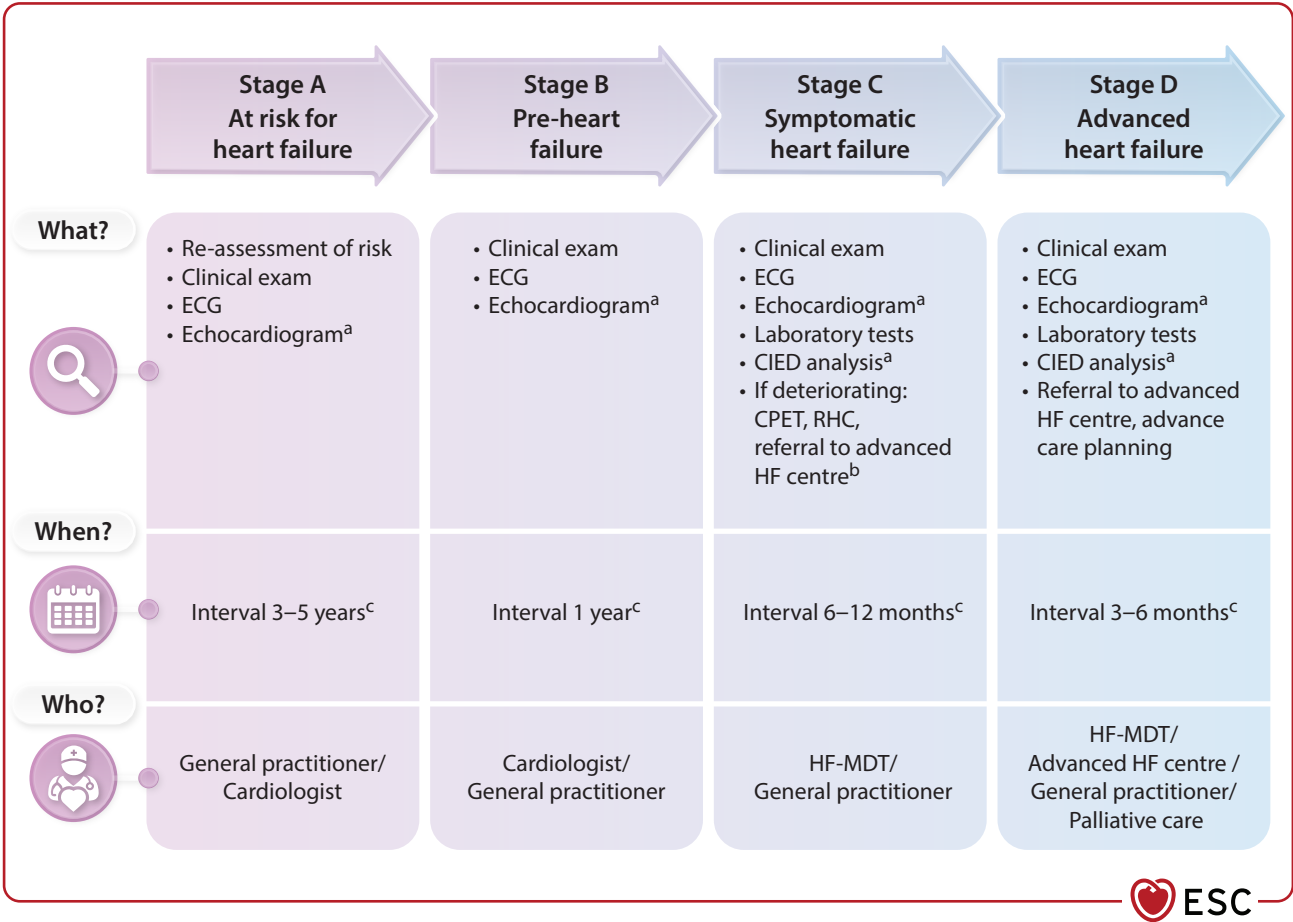


Figure 22 Long-term follow-up of patients with stable heart failure—what—when—who. CIED, cardiac implantable electronic device; CPET, cardiopulmonary exercise testing; ECG, electrocardiogram; HF, heart failure; HF-MDT, heart failure multidisciplinary team; RHC, right heart catheterization. ^aIf deemed clinically relevant. ^bCheck ‘I NEED HELP’ criteria, [Figure 16](#); [Table 15](#) (Section 8). ^cIntervals may be shorter.

status, liver, kidney, and thyroid function). What is needed, at which interval and who has to perform the follow-up assessment depend on HF stage ([Figure 22](#)). The necessity of cardiologist-led follow-up for these patients remains uncertain, with some studies suggesting that primary care follow-up may be appropriate.¹⁰⁵⁴ However, evidence-based intervention uptake is low in many settings, and studies indicate that follow-up by an HF clinic, alongside quality improvement registries, may lead to better therapy optimization and improved outcomes.^{990,994,1054–1057}

An ECG is recommended to detect QRS prolongation, which may qualify patients for CRT, as well as to identify conduction disturbances, AF, and other arrhythmias. Echocardiography should be performed if there is clinical deterioration or if otherwise deemed relevant. Trials investigating the use of biomarkers, such as BNP and NT-proBNP, to guide therapy in HFrEF have yielded inconsistent results.^{1058–1060} Although these biomarkers are strong prognostic indicators, it remains unclear what additional value a biomarker-guided strategy offers beyond strict adherence to FMT. As such, routine biomarker measurement for guidance on diuretic therapy is not currently supported by the evidence.

11.6.2. Advance care planning and palliative care

While the disease trajectory of each patient with HF is unique, there is a generalizable pattern of gradual decline, punctuated by episodes of acute deterioration leading to death. Communication about disease trajectory and advance care planning should be incorporated by all healthcare professionals throughout the course of the disease.^{1061,1062}

Advance care planning is a structured process that facilitates the exploration and communication of the patient’s personal values, life goals, and preferences regarding future medical care, and may include preferences for place of death and resuscitation (which may include deactivating devices, such as ICD or durable MCS). Advance care planning should be documented, regularly reviewed, and routinely communicated to all those involved in the patient’s care. Evidence shows that proactive advance care planning can reduce hospitalizations and improve QoL and end-of-life care, yet many patients with HF lack formalized plans, highlighting the need for better implementation.⁶²⁷

Palliative care, defined as patient- and family-centred care aimed at optimizing health-related QoL by anticipating, preventing, and alleviating suffering, should be integrated into the management of patients with HF in an advanced stage of the

disease. The role of palliative and supportive care spans the entire course of HF, beginning early in the disease, intensifying as the condition progresses, and extending into caregiver bereavement. Evidence supports the use of multidisciplinary palliative care in people with advanced HF to improve patient-reported outcomes (symptom burden, depression, functional status, QoL), but trials do not identify who would benefit most from specialist palliative referral.^{1063–1065} There are no sufficiently robust multicentre evaluation phase trials to provide generalizable findings.

Recommendation Table 26 — Recommendations for long-term follow-up (see [Evidence Tables 101 and 102](#))

Recommendations	Class ^a	Level ^b
Proactive discussion regarding HF trajectory, goals of care, and advance care planning is recommended in patients with advanced HF to facilitate communication on end of life and QoL. ⁶²⁷	I	B1
Access to an integrated HF palliative care multidisciplinary team is recommended for patients in an advanced HF stage to improve QoL and reduce symptom burden. ^{1066–1068}	I	B2

HF, heart failure; QoL, quality of life.

^aClass of recommendation.

^bLevel of evidence.

12. Specific conditions

12.1. Pregnancy and heart failure

Heart failure in pregnancy requires careful multidisciplinary management.^{1069–1071} The core components of HF care in pregnancy involves multiple aspects from pre-conception to postpartum (see [Supplementary data online, Table S26](#)).

Specific recommendations for pre-conception care and management of HF during pregnancy can be found in [Recommendation Table 27](#). For an overview of managing HF in pregnancy, see [Supplementary data online, Section S9.1](#). For a comprehensive and detailed guidance, we refer to the 2025 *ESC Guidelines for the management of cardiovascular disease and pregnancy*.⁷

Recommendation Table 27 — Recommendations for heart failure and pregnancy (see [Evidence Tables 103–107](#))

Recommendations	Class ^a	Level ^b
Pre-conception care and counselling is recommended for patients with HF to facilitate decision-making surrounding HF treatments, pregnancy, contraception, pre-implantation genetic screening, and assisted reproductive therapies.	I	C

Continued

In pregnant women with HFrEF, it is recommended that non-selective beta-blockers are switched to beta ₁ -selective blockers (metoprolol, bisoprolol) with close mother and foetus monitoring to reduce the risk of adverse foetal events and a lower birth weight. ^{1072,1073}	I	C
ACE-Is, ARBs, ARNIs, MRAs, ivabradine, and SGLT2-Is are not recommended during pregnancy due to the risk of foetotoxicity or teratogenicity. ^{1074–1076}	III	C

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ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor-neprilysin inhibitor; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; MRA, mineralocorticoid receptor antagonist; SGLT2-I, sodium-glucose co-transporter 2 inhibitor.

^aClass of recommendation.

^bLevel of evidence.

12.2. Cardiomyopathies

Cardiomyopathies are a common and specific cause of HF.¹²⁴ The 2023 *ESC Guidelines for the management of cardiomyopathies* described cardiomyopathy phenotypes based on structural (e.g. ventricular hypertrophy, dilatation, or scar) and functional (e.g. systolic or diastolic dysfunction) features.¹²⁴ Detailed recommendations and algorithms for the diagnosis and management of cardiomyopathies are provided in the 2023 *ESC Guidelines for the management of cardiomyopathies*.¹²⁴

12.2.1. Hypertrophic cardiomyopathy

Patients with HCM may develop HFrEF; in such cases, they should in general be treated as other patients with HFrEF. However, many patients develop symptoms of HF without having HFrEF, and these should be treated according to the cardiomyopathy guidelines (see [Supplementary data online, Section S9.2](#)).¹²⁴

12.3. Amyloidosis

12.3.1. Epidemiology and diagnosis

Cardiac amyloidosis is an infiltrative cardiomyopathy caused by amyloid fibril deposition in the extracellular space, resulting in increased myocardial thickness and stiffness, leading to a restrictive physiology.^{1077,1078} The two most prevalent forms of CA are immunoglobulin light chain CA (AL-CA) and transthyretin amyloidosis (ATTR). Transthyretin amyloidosis includes the wild-type (>90% of cases), and the hereditary or variant forms.¹⁰⁷⁷ Greater awareness of CA, streamlined diagnostic processes, and the emergence of new treatments has led to an increase in the prevalence of CA.^{1079,1080} The prevalence of CA is higher in males.¹⁰⁸¹

Cardiac amyloidosis should be suspected in patients with HF and increased LV wall thickness in the presence of specific red flags ([Table 20](#)), particularly in patients aged >65 years.

A timely, definitive diagnosis should be obtained as patient outcomes depend largely on early initiation of therapy (particularly in AL-CA). Cardiac amyloidosis can be diagnosed using both invasive and non-invasive diagnostic criteria. Invasive diagnostic criteria apply to all forms of CA, whereas non-invasive criteria are accepted only for transthyretin CA (ATTR-CA).¹⁰⁷⁷ [Figure 23](#) displays the diagnostic work-up and treatment options. For more diagnostic details, the Task Force refers to the 2021 ESC position statement on CA.¹⁰⁷⁷

Table 20 'Red flags' for most common forms of cardiac amyloidosis

Category	Red flag	TTR	AL
Extracardiac	Polyneuropathy	X	X
	Dysautonomia	X	X
	Skin bruising	–	X
	Macroglossia	–	X
	Deafness	X	–
	Bilateral carpal tunnel syndrome	X	–
	Ruptured biceps tendon	X	–
	Lumbar spinal stenosis	X	–
	Vitreous deposits	X ^a	–
	Family history	X ^a	–
	Kidney insufficiency	X	X
	Proteinuria	X	X
Cardiac clinical	Hypotension or normotension if previously hypertensive	X	X
ECG	Pseudo-infarct ECG pattern	X	X
	Low/decreased QRS voltage relative to LV thickness	X	X
	AV conduction disease	X	X
Laboratory	Disproportionally elevated NT-proBNP relative to HF severity	X	X
	Persistently elevated cardiac troponin levels	X	X
Echocardiography	Granular sparkling of myocardium	X	X
	Increased right ventricular wall thickness	X	X
	Increased valve thickness	X	X
	Increased interatrial septum thickness	X	X
	Pericardial effusion	X	X
	Reduced longitudinal strain with apical sparing pattern	X	X
CMR ^b	Global subendocardial or transmural LGE	X	X
	Elevated native T1 values	X	X
	Increased extracellular volume	X	X
	Abnormal gadolinium kinetics	X	X

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AL, immunoglobulin light chain; CA, cardiac amyloidosis; CMR, cardiac magnetic resonance; ECG, electrocardiogram; HF, heart failure; LGE, late gadolinium enhancement; LV, left ventricular; NT-proBNP, N-terminal-pro-B-type natriuretic peptide; TTR, transthyretin.

^aHereditary TTR-CA.

^bCMR can be diagnostic for CA.

12.3.2. Disease-specific treatments

12.3.2.1. Immunoglobulin light chain cardiac amyloidosis

Treatment of AL-CA includes chemotherapy or autologous stem cell transplantation with careful monitoring of haematological response, cardiac and organ function. Management of AL-CA should be undertaken by a multidisciplinary team involving

oncohaematology specialists and cardiologists, and whenever possible, patients should be referred to specialized centres.

12.3.2.2. Transthyretin cardiac amyloidosis

Therapeutic agents for ATTR-CA include drugs that inhibit hepatic synthesis of transthyretin (TTR), stabilize the tetramer, or disrupt the amyloid fibrils. Based on the results of large RCTs, the Task Force recommends treatment with a TTR stabilizer or a TTR silencer with proven safety and efficacy on morbidity and mortality, in patients with either wild-type or hereditary ATTR-CA and a clinical history of HF with NYHA classes I–III. These drugs include the TTR stabilizers tafamidis and acoramidis and the TTR silencer vutrisiran.^{1082–1084}

In the ATTR-ACT trial, tafamidis reduced all-cause death and CV hospitalizations over 30 months in both hereditary and wild-type ATTR-CA.¹⁰⁸² Benefits on functional capacity occurred within 6 months, whereas reduction in all-cause death was observed from 18 months onwards. It showed limited impact in advanced cases (NYHA class III).

In the ATTRIBUTE-CM trial, after 30 months of treatment, acoramidis was superior to placebo in a hierarchical analysis (all-cause death, CV hospitalization, change from baseline in NT-proBNP, and change from baseline in 6-min walk test).¹⁰⁸³ Although it did not independently lower death at 30 months, it did reduce the combined endpoint of all-cause death or CV hospitalizations. Notably, survival rates in the placebo arm were higher than in the treatment group of ATTR-ACT, possibly reflecting earlier disease stage and the improvements in earlier diagnosis and access to specialist care. Furthermore, in the open-label extension study, early initiation and continuous use of acoramidis significantly reduced death at 42 months, as compared with the later initiation of acoramidis.¹⁰⁸⁵

In the APOLLO-B trial, patisiran (i.v.) reduced functional decline over 12 months.¹⁰⁸⁶ However, no data are available regarding morbidity and mortality. In the HELIOS-B trial, vutrisiran (subcutaneous) reduced the risk of the combined endpoint of all-cause death and recurrent CV events vs placebo.¹⁰⁸⁴ Also, the risk of all-cause death through 42 months was lowered with vutrisiran. These benefits persisted regardless of background tafamidis use at baseline (~40% of the trial population).¹⁰⁸⁴

None of the studies provided a head-to-head comparison of TTR silencers vs stabilizers nor a head-to-head comparison of monotherapy vs combination therapies. Several studies are ongoing with further therapeutic options acting on gene editing, RNA silencing, or the repletion of amyloid deposits, and might change the management of ATTR-CA in the future.

12.3.3. Treatment of heart failure and cardiovascular comorbidities

Evidence for the use of FMT in patients with CA is still scarce and limited to post-hoc analyses of RCTs or retrospective studies.^{1087–1092} Patients with CA may poorly tolerate standard HF therapies due to restrictive physiology, autonomic dysfunction, atrioventricular conduction disturbance, and kidney impairment. Diuretics remain central for congestion management but must be used cautiously within a narrow euvolaemic window.^{1089,1090} Increasing diuretic dose is a signal of disease progression with worse outcomes.^{1093,1094} Evidence supports the safety and benefit of SGLT2-Is.^{73,1089–1092} MRA might also be useful.¹⁰⁹²

Beta-blockers and ACE-I/ARNI/ARB use requires caution, individualized decisions, and close monitoring.

In patients with CA and AF, anticoagulation is indicated regardless of CHA₂DS₂-VA score, due to high thromboembolic risk.^{1077,1095–1097} Transoesophageal echocardiography is necessary before cardioversion to exclude intracardiac thrombus, regardless of the duration of anticoagulation before the procedure.¹⁰⁹⁸ The use of digoxin is generally discouraged due to high risk of toxicity. Few patients with CA and advanced symptoms may be eligible for heart transplantation in line with international criteria.⁵⁸⁰

Recommendation Table 28 – Recommendations for cardiac amyloidosis (see Evidence Tables 108–111)

Recommendation	Class ^a	Level ^b
Treatment with a TTR silencer (vutrisiran) or stabilizer (tafamidis or acoramidis) for patients with variant or wild-type transthyretin cardiac amyloidosis and NYHA classes I–III is recommended to reduce progression of symptoms and the risk of CV hospitalization and all-cause death. ^{1082–1084}	I	A

CV, cardiovascular; NYHA, New York Heart Association; TTR, transthyretin.
^aClass of recommendation.
^bLevel of evidence.

12.4. Myocarditis

12.4.1. Epidemiology and diagnosis

The contribution of myocarditis as a cause of HF is estimated to be less than 5%.¹⁰⁹⁹ When patients with myocarditis present with DHF, LVEF <40%, life-threatening arrhythmias (ventricular tachycardia/fibrillation, third-degree atrioventricular block), and/or extensive LGE on CMR, they should be viewed as high risk.^{1100,1101} High-risk patients may require an endomyocardial biopsy in order to tailor treatment,⁸ whereas lower-risk patients may be examined with CMR imaging. The clinical presentation ranges from impaired LV function to refractory HF, CS (fulminant myocarditis), or sudden cardiac arrest. Some patients present primarily with arrhythmias. Prognosis depends on initial presentation.^{1100,1101} Patients with fulminant myocarditis should be transferred or discussed with a center that has the possibility to perform endomyocardial biopsy and provide MCS support. High-risk patients have a high mortality and may need temporary MCS and/or in the long term heart transplantation or durable MCS.^{1102–1104}

12.4.2. Treatment

The treatment algorithm is reported in the 2025 ESC Guidelines for the management of myocarditis and pericarditis.⁸ Patients with myocarditis and HFrEF should be empirically treated with FMT for HFrEF (see Section 7). Routine immunosuppression is not recommended in patients with preserved LVEF and should be

reserved for fulminant, non-infectious forms of myocarditis to stabilize the patients.⁸

12.5. Adult congenital heart disease

Adult congenital heart disease is a heterogeneous group of structural heart abnormalities varying in complexity. There has been an improvement in survival with contemporary care.¹¹⁰⁵ Consequently, HF in ACHD is increasingly common,^{1106,1107} and represents a major contributor to morbidity and mortality, accounting for 26%–46% of deaths, with higher rates seen in more complex lesions.^{1108–1110}

Managing HF in ACHD poses specific challenges related to original anatomy, presence of haemodynamic lesions, previous interventions, and extracardiac complications such as kidney and liver dysfunction.^{1111,1112}

Specific indications on the management of these patients are reported in the 2020 ESC Guidelines for the management of adult congenital heart disease¹¹¹¹ and in a recent ESC clinical consensus statement.¹¹¹³

13. Key messages

New classifications

- Heart failure has been redefined with two distinct phenotypes:
 - HFrEF: LVEF <50% + symptoms and/or signs of HF.
 - HFpEF: LVEF ≥50% + symptoms and/or signs of HF + objective evidence of cardiac structural and/or functional abnormalities.
- Stages of HF (A–D) have been adopted:
 - Stage A: individuals at risk for HF but without signs, symptoms, and cardiac abnormalities.
 - Stage B: pre-HF, defined as the presence of cardiac abnormalities without signs and symptoms of HF.
 - Stage C: symptomatic HF, defined as the presence of cardiac abnormalities and current or prior signs and/or symptoms of HF.
 - Stage D: advanced HF.
- New classifications for medical and interventional HF therapies have been introduced. The terms are designed to be dynamic and may therefore remain contemporary over time:
 - FMT—Foundational medical therapy: medical therapies for unselected patients with HF with Class I recommendation for reducing HFH and/or death.
 - AMT—Additional medical therapy: medical therapies with Class IIa/IIb recommendations or Class I recommendations for improving symptoms/QoL or reducing morbidity or mortality in specific subsets of patients with HF.
 - GDIT—Guideline-directed interventional therapy: all guideline-recommended implantable devices or interventional therapies.

Prevention of heart failure

- A strict control of HF risk factors is recommended to prevent HF.

Diagnosis of heart failure

- Measurement of natriuretic peptides and echocardiography have key roles in the diagnosis of HF. Additional testing is useful for aetiologies and phenotypes.
- Multimodality aetiology assessment should be sought to find potential treatable causes of HF.

Guideline-recommended therapies

- ARNIs or ACE-Is (ARB in case of intolerance) beta-blockers, MRAs, and SGLT2-Is are recommended as FMT for patients with HFrEF, whereas MRAs and SGLT2-Is are recommended as FMT for patients with HFpEF.
- ICDs are recommended in selected patients with HFrEF, LVEF $\leq 35\%$, and an ischaemic aetiology, and should be considered in those with a non-ischaemic aetiology.
- CRT-P or CRT-D is recommended in patients with HFrEF, LVEF $\leq 35\%$, in sinus rhythm, with an LBBB ≥ 150 ms and should be considered in those with an LBBB ≥ 130 – 149 ms or non-LBBB ≥ 150 ms.

Decompensated heart failure

- Decompensated heart failure refers to acute or gradual onset of symptoms and/or signs of HF, severe enough to deserve urgent medical attention, leading to an unplanned hospital admission or an emergency department visit or an ambulatory visit, and requiring subsequent initiation or intensification of treatment.
- Treatment of DHF is based on addressing any associated triggers urgently, diuretics for congestion with the addition of vasodilators for high blood pressure, inotropes, and/or temporary MCS for hypoperfusion.
- Patients hospitalized for HF should be carefully evaluated to exclude persistent signs of congestion. FMT must be started or optimized before discharge and close visits in the early post-discharge period must be planned.

Advanced heart failure

- Patients with advanced HF refractory to FMT and GDIT who do not have absolute contraindications must have timely referral for consideration of heart transplantation or MCS.

Cardiovascular comorbidities

- Catheter ablation should be considered only in carefully selected patients with HFrEF and symptomatic AF. Selection criteria include high-burden AF, less than 1-year duration of continuous persistent AF, and a clear cause–effect relationship between AF and HF.
- Patients with obstructive CAD and HF need to be treated conservatively or invasively (CABG or PCI) according to clinical and anatomical suitability for revascularization, HF phenotype and history, presence of angina, comorbidities, frailty, and patient preference.
- Surgical aortic valve replacement or TAVI, as assessed by the Heart Team, is recommended in patient with severe low-flow

low-gradient aortic stenosis and HF regardless of LVEF and in patients with severe low-flow low-gradient aortic stenosis and HFrEF.

- Mitral TEER is recommended in selected patients with severe ventricular secondary MR and HFrEF after optimization of FMT and CRT implantation (if needed).

Non-cardiovascular comorbidities

- Patients should be routinely screened and treated for non-CV comorbidities (e.g. diabetes, CKD, obesity, ID, anxiety and depression) and frailty.
- Lifestyle interventions, exercise, diet, and patient education are fundamental steps for body-weight management in patients with HF and obesity, followed by pharmacological and surgical interventions.
- It is recommended to switch glucose-lowering treatment from agents without proven CV efficacy or safety to agents with proven CV benefit in patients with HF and diabetes.

Multidisciplinary management

- A multidisciplinary HF management programme is recommended.
- Cardiac rehabilitation, with exercise training, is a key tool to reduce hospitalizations and improve functional capacity and QoL in patients with HF.
- Remote haemodynamic monitoring of PA pressure should be considered in selected patients; non-invasive telemonitoring modalities may be considered.

Specific conditions

- Pre-conception care and counselling is recommended for patients with stage B or C HF to facilitate decision-making surrounding pregnancy.
- Specific disease-modifying therapies are recommended in patients with CA.

14. Gaps in evidence

Despite advances in the management of HF, multiple areas of uncertainty remain. Future research should address the following gaps.

(1) Definitions and prevention

- Consensus regarding normal values/ranges of ejection fraction.
- More information on the incidence, prevalence and prognostic importance of HF with improved LVEF, and a need to validate and evaluate the clinical importance of the new definition proposed by the Second Universal Definition of HF.⁴⁷
- The benefits and risks of various weight-loss methods for HF risk reduction in obesity need further studies.
- More research is needed to understand if SGLT2-Is and MRAs prevent HF in patients without diabetes but with comorbidities like obesity, hypertension, and/or CKD.
- Studies are needed to better understand if CPAP therapy lowers risk of HF in patients with sleep apnoea.

(2) Diagnosis and phenotyping

- Further research into the underlying characteristics, pathophysiology, diagnosis, and phenotyping of HFpEF.
- Role of artificial intelligence in the diagnostic pathway for HF as well as for patient inclusion in RCTs.
- Validated diagnostic protocols including deeper phenotyping for the diagnosis of HFpEF.
- Validation of the historic exercise-induced pulmonary capillary wedge pressure cut-offs used in the diagnosis of HFpEF.
- Prospective validation of age-related natriuretic peptide cut-offs in chronic HF.
- Degree of influence of BMI on natriuretic peptides to allow for individual adjustment of expected ranges.
- Prospective studies to test the diagnostic yield for endomyocardial biopsy, and whether a biopsy would affect treatment decisions and outcomes in patients with unexplained cardiomyopathy and evidence of inflammation/infiltration/storage on CMR or CT-positron emission tomography.

(3) Medical treatment and devices

- Contemporary reproducibility of old trials on medical and device therapies.
- Future trials on medical/device therapy for HF should adhere to contemporary FMT.
- Randomized clinical trials comparing different MRAs in both HFrEF and HFpEF.
- Randomized clinical trials investigating the optimal medical therapy for HFpEF.
- Randomized clinical trials comparing different GLP-1 RAs.
- Randomized clinical trials on the optimal sequencing when initiating FMT both in HFrEF and HFpEF.
- More data on the role of potassium binders, both in patients with and without CKD.
- Management strategies and therapies for HF with improved LVEF.
- Indications for ICDs in specific subgroups of patients, and optimal selection of ICD candidates in HFrEF, including patients with ischaemic and non-ischaemic HF.
- Efficacy of CRT in AF.
- Randomized controlled trials with morbidity and mortality endpoints and long-term follow-up, comparing CSP and trans-coronary vein CRT in patients with HFrEF.
- Randomized controlled trials with morbidity and mortality and long-term follow-up on interatrial shunt devices in patients with HFrEF and HFpEF.

(4) Decompensated heart failure

- Evidence on medical therapy with prognostic benefit in DHF.
- Evidence for best decongestion protocols in patients with and without CKD and in the setting of continued/initiation of FMT.
- Evidence-based use of imaging techniques and biomarkers that have an impact on patients' clinical course in DHF.
- Definition of evidence-based treatment options and therapeutic algorithms for patients with CS.
- Better definition of the role of MCS in CS.

- Better definition of the role of inotropes, vasopressors and vasodilators in DHF.
- Studies on the best early post-discharge risk-based management protocols to reduce early readmission to hospital.

(5) Advanced heart failure

- Benefit of FMT, AMT, and GDIT in advanced HFrEF needs further investigation.
- Which medication to down-titrate/stop first in patients with advanced HF who no longer tolerate all FMT.
- Durable MCS (LVADs) vs medical treatment in INTERMACS >4 without high-risk features.
- Optimal lifetime management strategies in very young adults with advanced HF.
- Advances in durable MCS, including strategies to reduce the risk of infection.
- Advances in medical treatment for the many patients who cannot undergo MCS or heart transplantation, including development of treatment strategies, and novel inotropes or myotropes for patients with advanced HF.

(6) Cardiovascular comorbidities

- Adequately powered RCTs showing the best strategy for management of AF in patients with HF.
- Selection of patients with HFrEF and HFpEF that may benefit from AF ablation.
- Better characterization of patients with arrhythmia-associated HF, especially patients with AF-associated HF.
- Randomized controlled trials investigating best management options for patients with AF-associated HF, both in terms of HF management and management of AF.
- Randomized controlled trials to assess the role of coronary revascularization and different revascularization strategies in patients with HFrEF and HFpEF on current FMT.
- Longer-term follow-up results for mitral TEER interventions in HFrEF.
- Role of mitral TEER in patients with HFrEF who do not fulfil specific echocardiographic and clinical criteria.
- Treatment of atrial secondary MR in patients with HFpEF.
- Further RCTs to assess the role of transcatheter valve intervention in patients with HF and TR.

(7) Non-cardiovascular comorbidities

- More clinically appropriate definition of ID in HF.
- Further RCTs, using the novel definition of ID as an inclusion criterion, to assess the prognostic role of i.v. iron supplementation.
- The prognostic role of HF treatment in patients with severe kidney dysfunction (i.e. eGFR <30 mL/min/1.73 m²), excluded from RCTs.
- The effects of GLP-1 RAs and tirzepatide on morbidity and mortality in patients with HFpEF and HFrEF.
- Treatment of SDB, both central and obstructive, needs further investigation to clarify benefit and safety concerns.

(8) Multidisciplinary management

- Studies on optimal models for follow-up of patients with stable HF.

- Long-term trials on exercise in patients with HF.
 - Studies to determine personalized approach for palliative care to improve patient-centred outcomes.
 - Models of care that incorporate contemporary patients with HF and the wider multidisciplinary team including cardio-kidney-metabolic professionals in HF care.
 - Patient selection for telemonitoring.
- (9) Specific conditions
- Randomized clinical trials for the treatment of peripartum cardiomyopathy.
 - Head-to-head comparison between different disease-modifying drugs in ATTR-CA.
 - Evidence for the combined use of disease-modifying drugs in ATTR-CA.
 - Evidence for gene editing and antibodies for the removal of amyloid deposits.
 - Efficacy of traditional HF therapies in patients with CA.
 - Identification of patients with ATTR-CA who could benefit from device therapy (e.g. ICD/CRT).
 - Data collection to support advice on screening for *TTR* gene carriers—when to initiate and follow-up interval.

15. Evidence tables

Evidence tables are available at [European Heart Journal](#) online.

16. Data availability

No new data were generated or analysed in support of this research.

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18. Appendix

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